

a	BC	00	01	11	10
A					
0					
1			1	1	d

$$a(ABC) = \sum \{4, 5\}$$

$$a = L_1$$

$$= A$$

b	BC	00	01	11	10
A					
0			1	1	1
1			1	1	d

$$b(ABC) = \sum \{0, 1, 3, 4, 5\}$$

$$b = L_1 + L_2$$

$$= \bar{B} + C$$

c	BC	00	01	11	10
A					
0			1	1	1
1			1	1	d

$$c(ABC) = \sum \{0, 1, 3, 4, 5\}$$

$$c = L_1 + L_2$$

$$= \bar{B} + C$$

d	BC	00	01	11	10
A					
0		1	1	1	1
1		1	1	d	d

$$d(ABC) = \sum \{0, 1, 2, 3, 4, 5\}$$

$$d = VCC$$

e	BC	00	01	11	10
A					
0			1	1	1
1				1	d

$$e(ABC) = \sum \{0, 1, 2, 5\}$$

$$e = L_1 + L_2 + L_3$$

$$= \bar{A}\bar{B} + \bar{B}C + \bar{A}\bar{C}$$

$$= \bar{A}(\bar{B} + \bar{C}) + \bar{B}\bar{C}$$

f	BC	00	01	11	10
A					
0			1	1	1
1			1	d	d

$$f(ABC) = \sum \{1, 2, 3, 5\}$$

$$f = L_1 + L_2$$

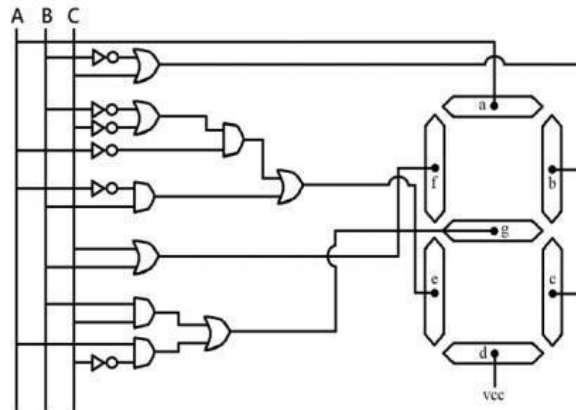
$$= B + C$$

g	BC	00	01	11	10
A					
0				1	
1		1		d	d

$$g(ABC) = \sum \{3, 4\}$$

$$g = L_1 + L_2$$

$$= A\bar{C} + BC$$



4. Resources:

- Personal Computer
 - Internet connection
- Simulation Tools:
- TinkerCAD
 - Logicly

5. Procedure:

1. Analyze the given problem and determine the requirements (number of inputs and number of outputs) of the problem.
2. Construct the truth table.
3. Simplify the output function using K-Mapping. Show this as your solution.
4. Draw the circuit diagram.
5. Construct the circuit and simulate its operation in Logic.ly by verifying the input / output relationship.
6. Simulate the operation in TinkerCAD by connecting the inputs, wires, electronic components, and output correctly following the pin configuration of the ICs.
7. After the pasting an output screenshot and writing down the link for the uploaded video recording below, write the conclusion, and answer the questions on section 8.

Problem No. 1

Design a combinational logic circuit with input (switch) and output (seven-segment display) that satisfy the condition or state diagram below:

When INPUT = 0, Display = 0 ◇ 2 ◇ 4 ◇ 6 ◇ 0 (Even Numbers)

When INPUT = 1, Display = 1 $\diamond$ 3 $\diamond$ 5 $\diamond$ 7 $\diamond$ 1 (Odd Numbers)

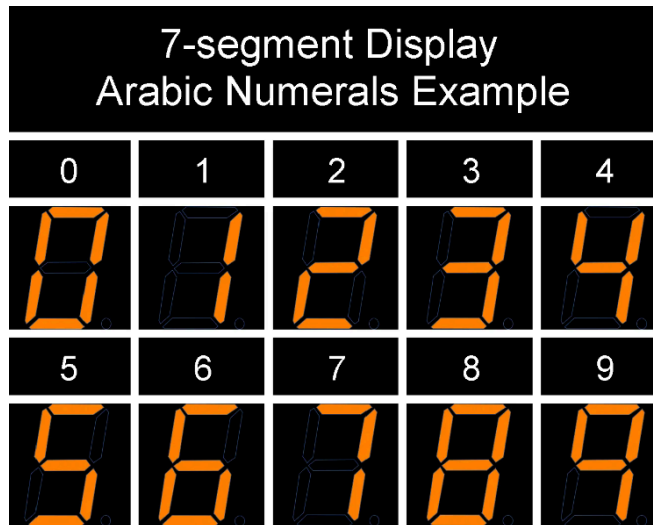


Image Source: <https://www.opledtw.com/blog/7-segment-16-segment-represent-numbers-letters-explained/>

TRUTH TABLE:

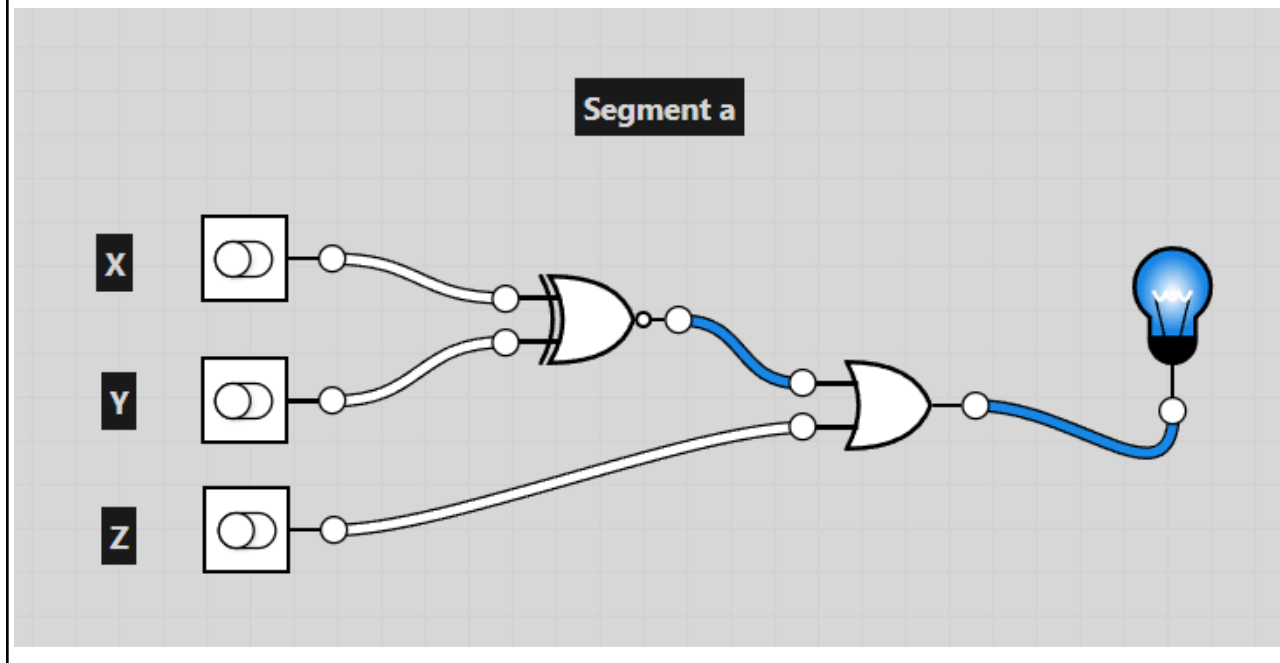
X(E/O)	Y	Z	a	b	c	d	e	f	g
0	0	0	1	1	1	1	1	1	0
0	0	1	1	1	0	1	1	0	1
0	1	0	0	1	1	0	0	1	1
0	1	1	1	0	1	1	1	1	1
1	0	0	0	1	1	0	0	0	0
1	0	1	1	1	1	1	0	0	1
1	1	0	1	0	1	1	0	1	1
1	1	1	1	1	1	0	0	0	0

SEGMENT a:

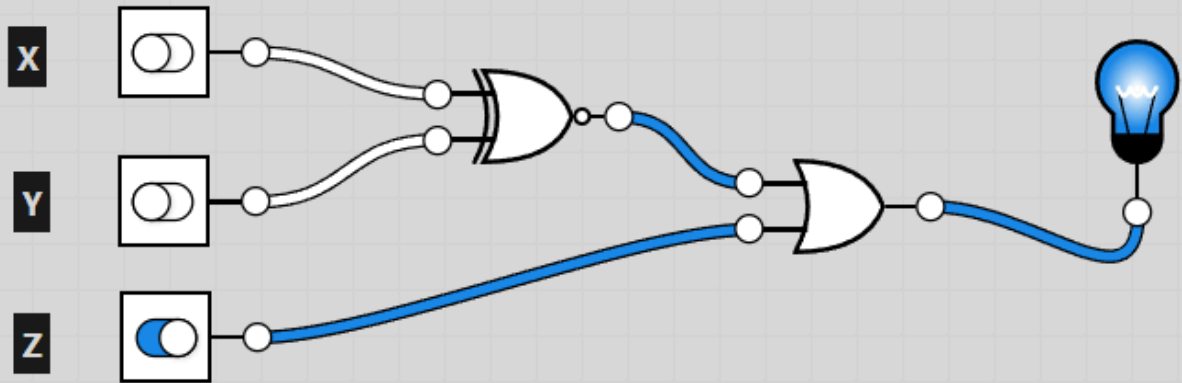
K-map:

X \ YZ	00	01	11	10
0	1	1	1	0
1	0	1	1	1

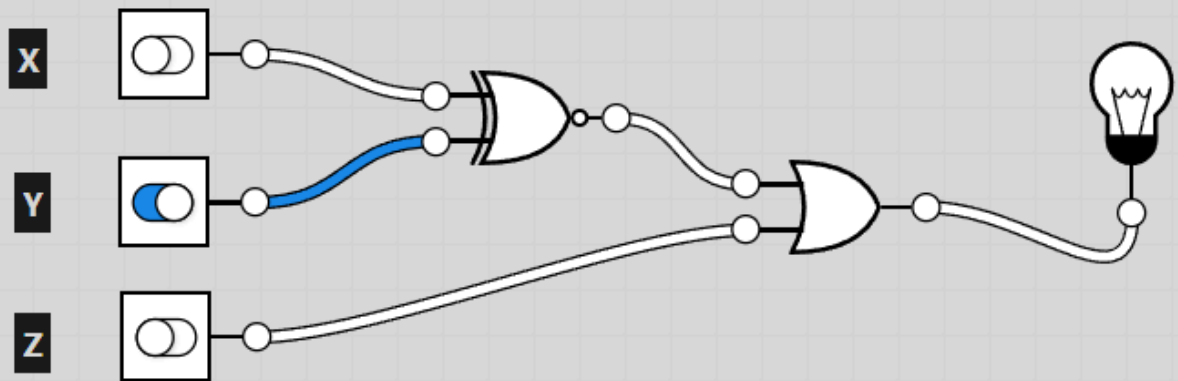
Logically:



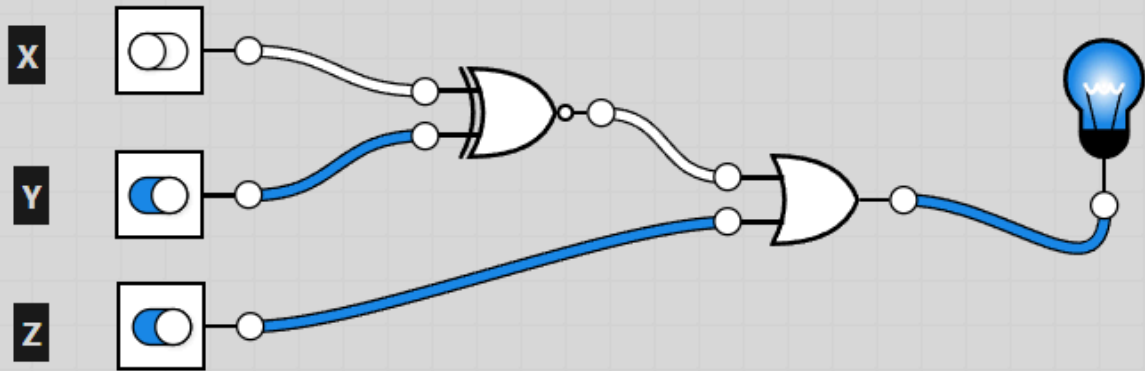
Segment a



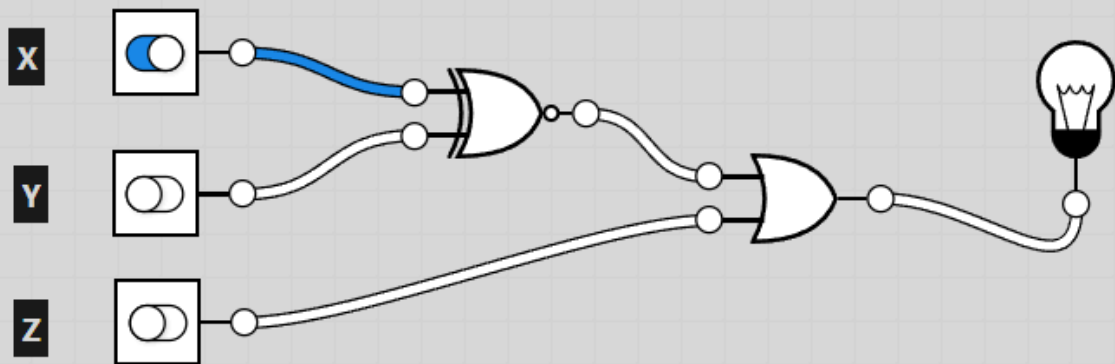
Segment a



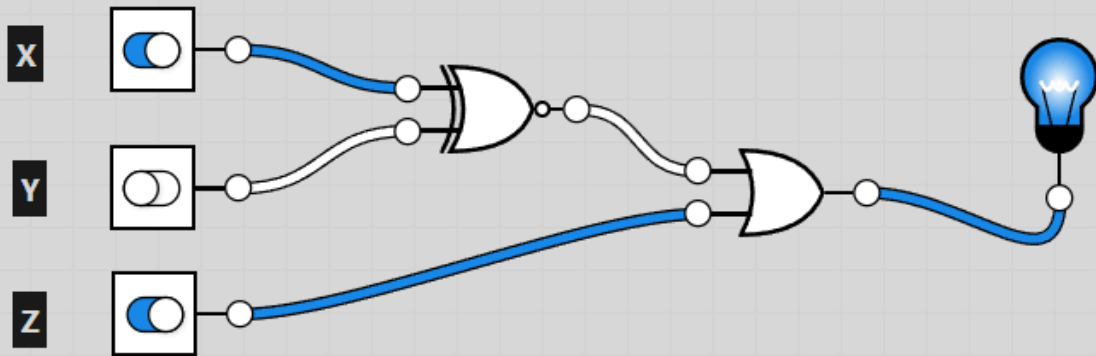
Segment a



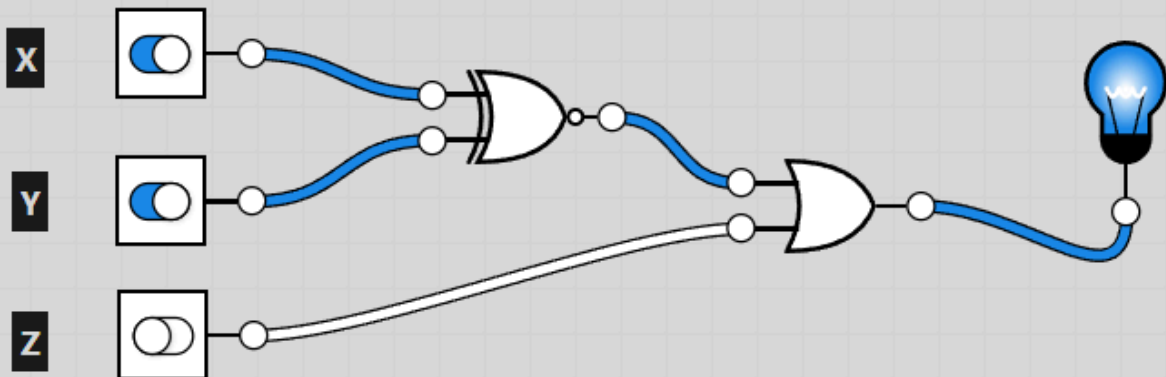
Segment a



Segment a



Segment a



Segment a

Boolean Expression: $a = Z + (X \text{ XNOR } Y)$

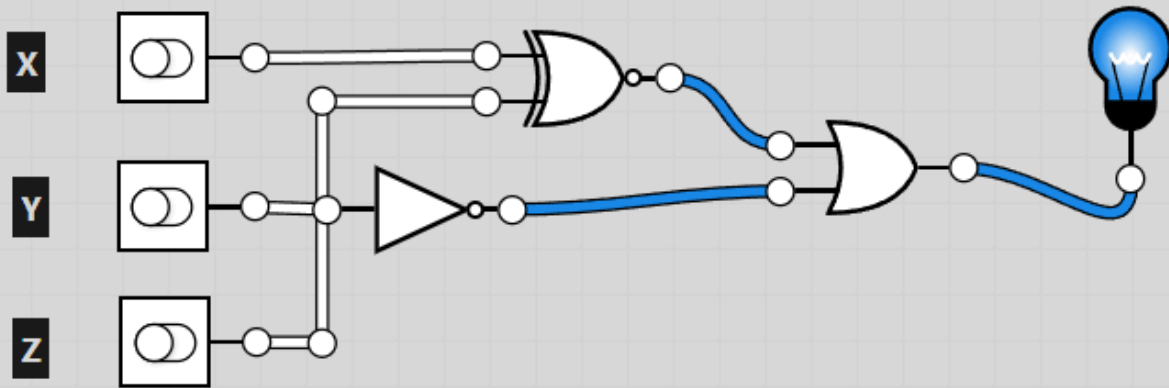
SEGMENT b:

K-map:

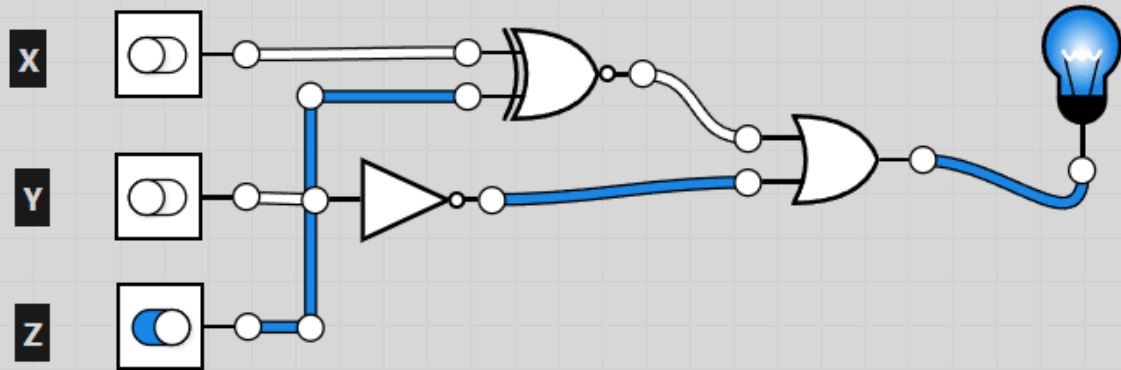
$X \setminus YZ$	00	01	11	10
0	1	1	0	1
1	1	1	1	0

Logically:

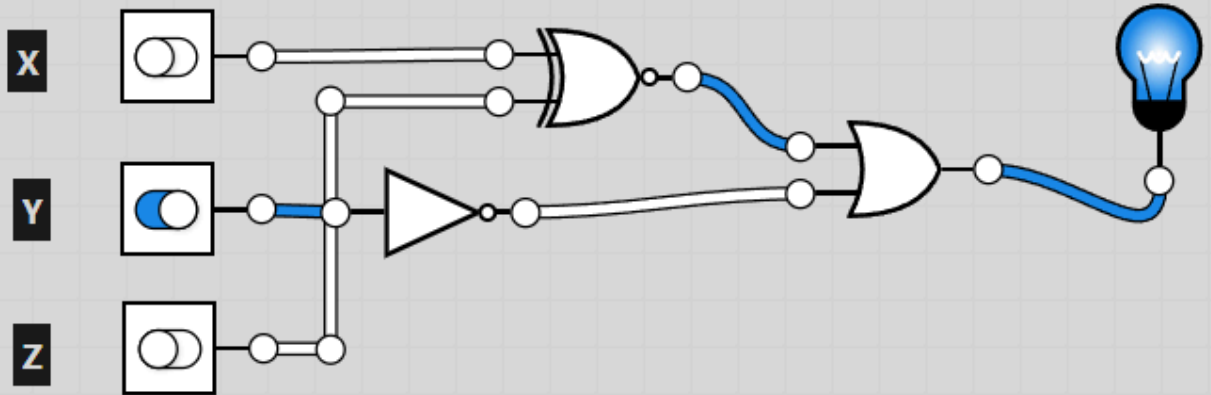
Segment b



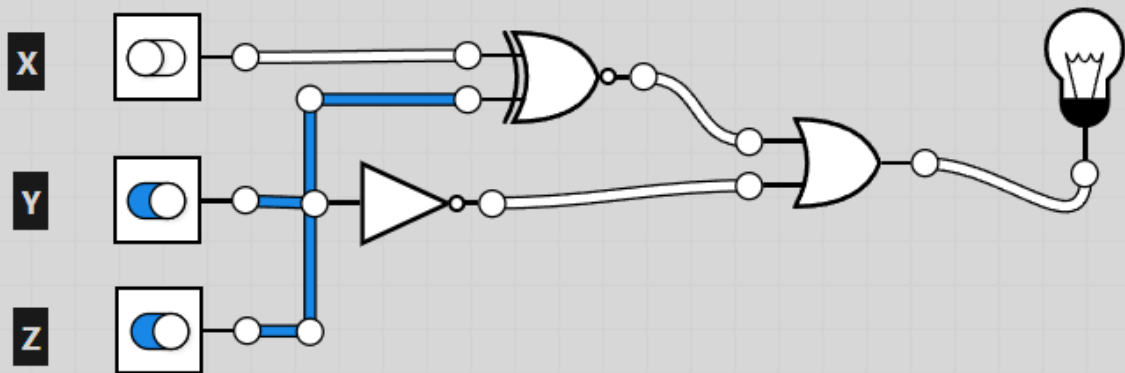
Segment b



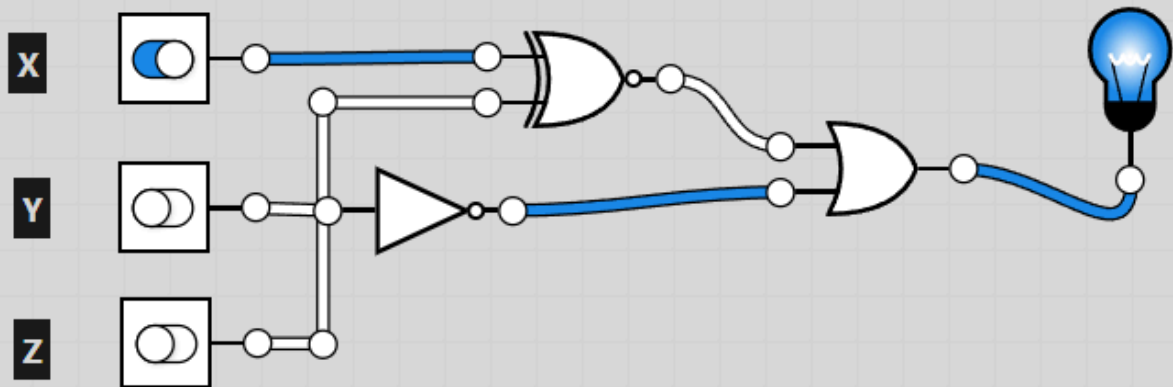
Segment b



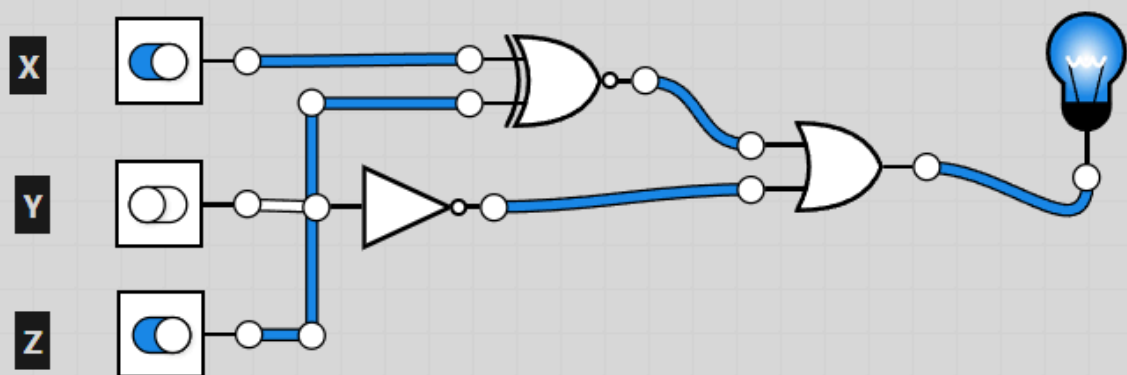
Segment b

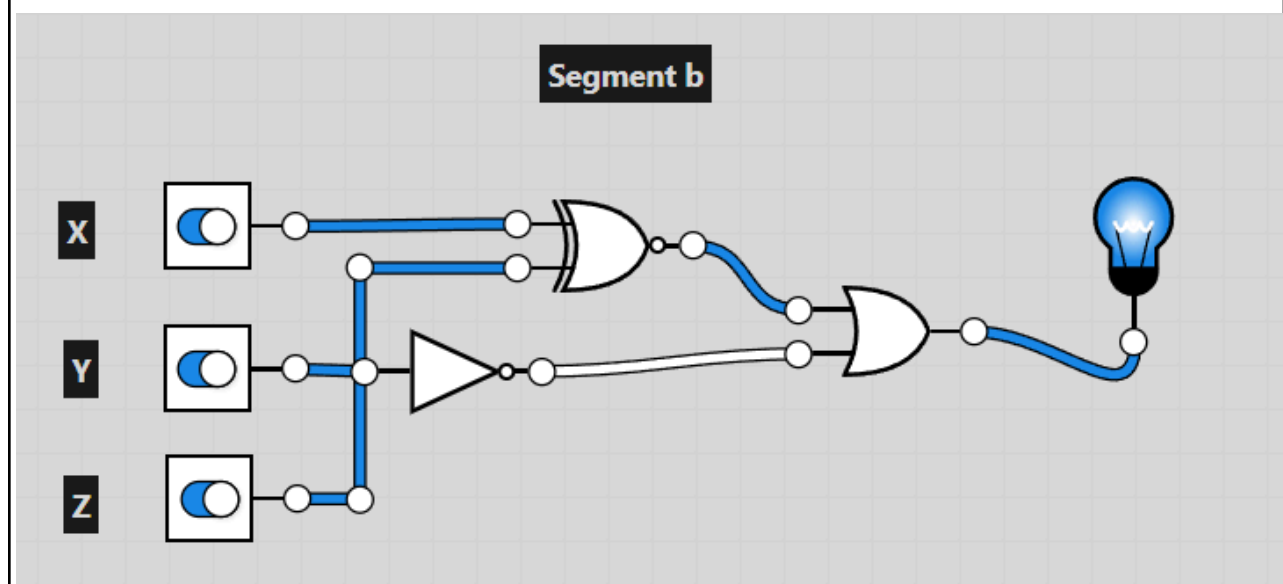
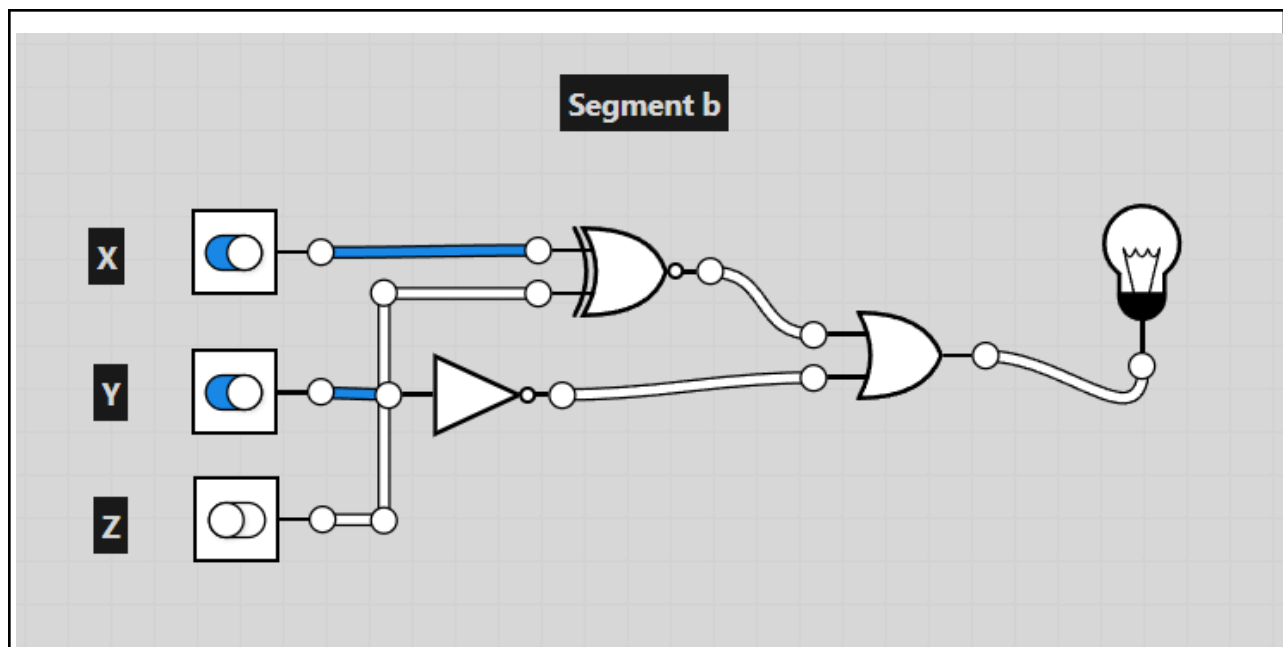


Segment b



Segment b





Boolean Expression: $b = Y' + (X \text{ XNOR } Z)$

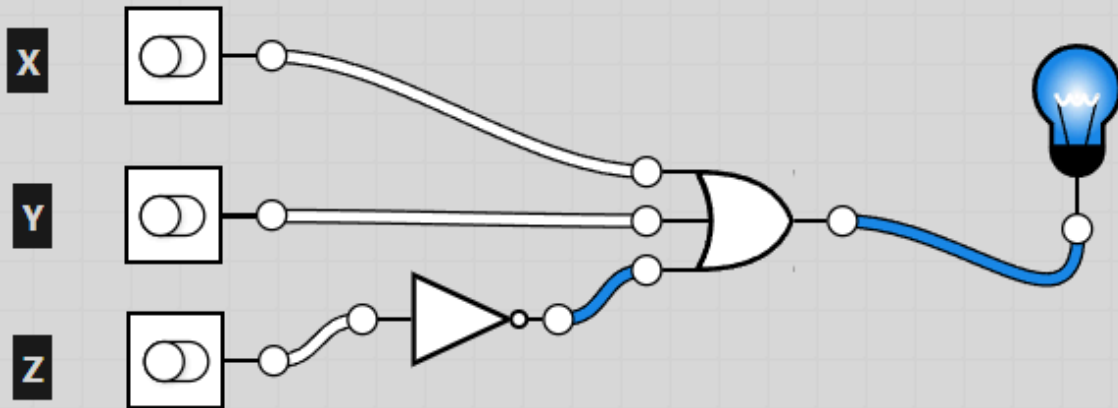
SEGMENT c:

K-map:

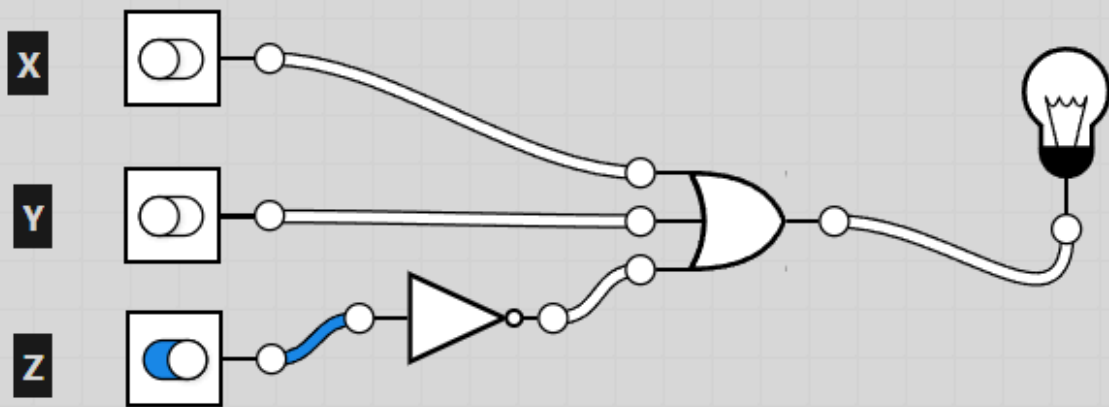
X \ YZ	00	01	11	10
0	1	0	1	1
1	1	1	1	1

Logically:

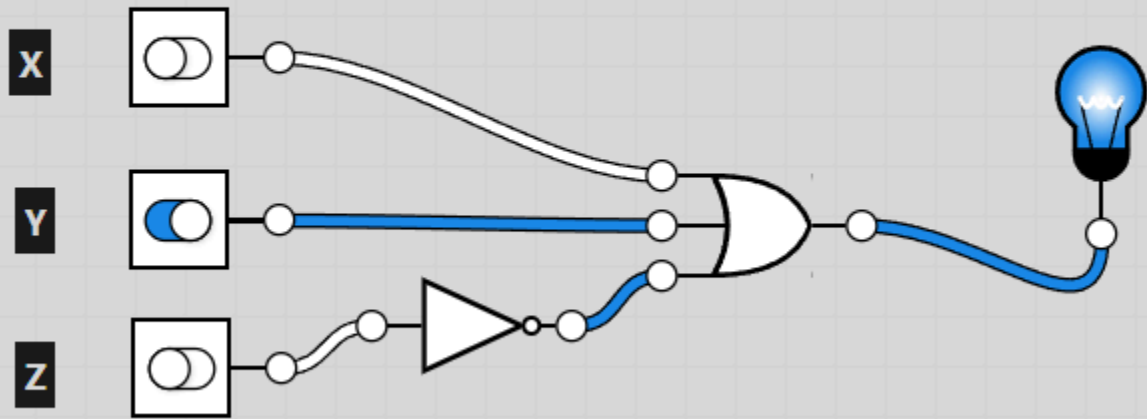
Segment c



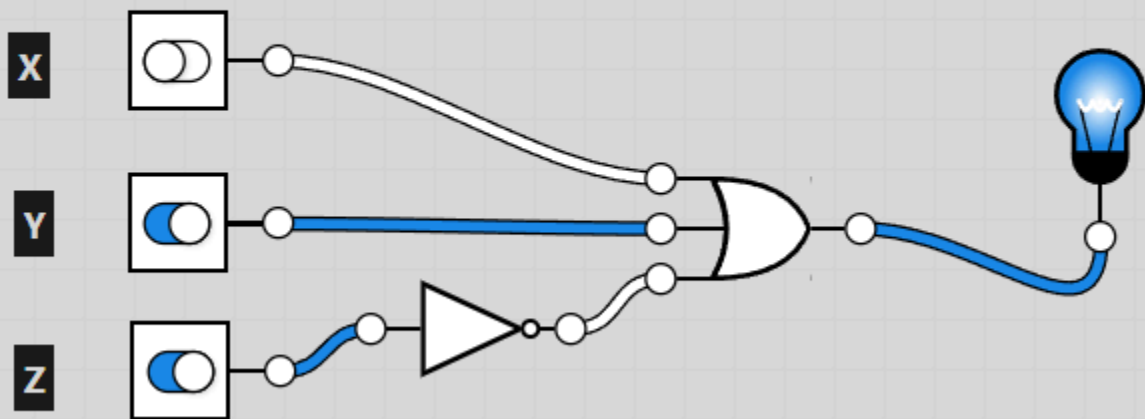
Segment c



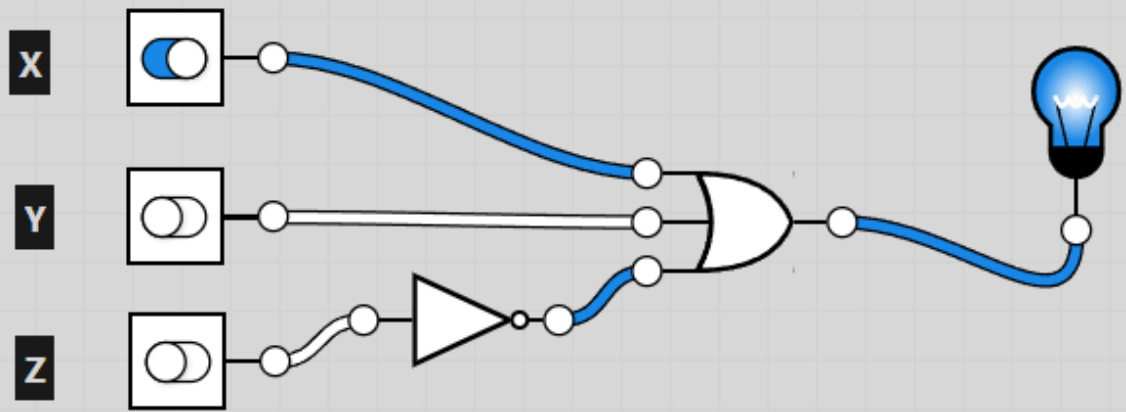
Segment c



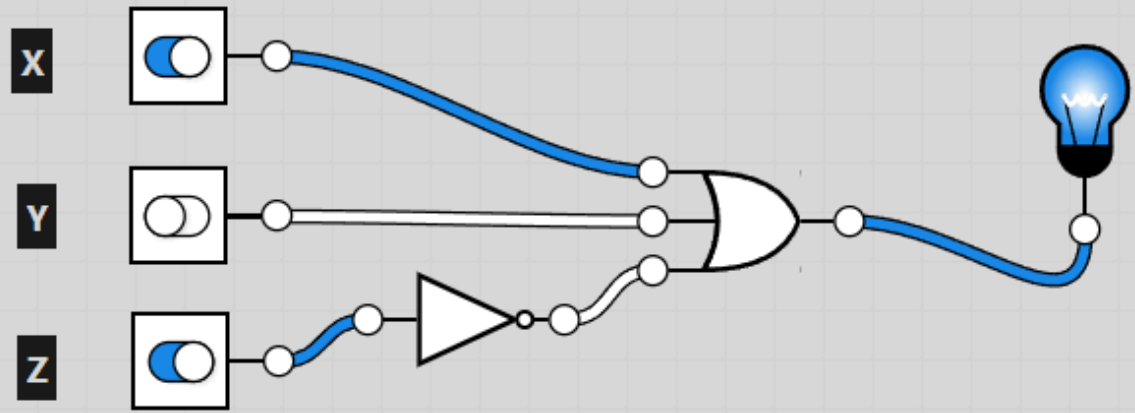
Segment c



Segment c



Segment c



Segment c

Segment c

Boolean expression: $c = X + Z' + Y$

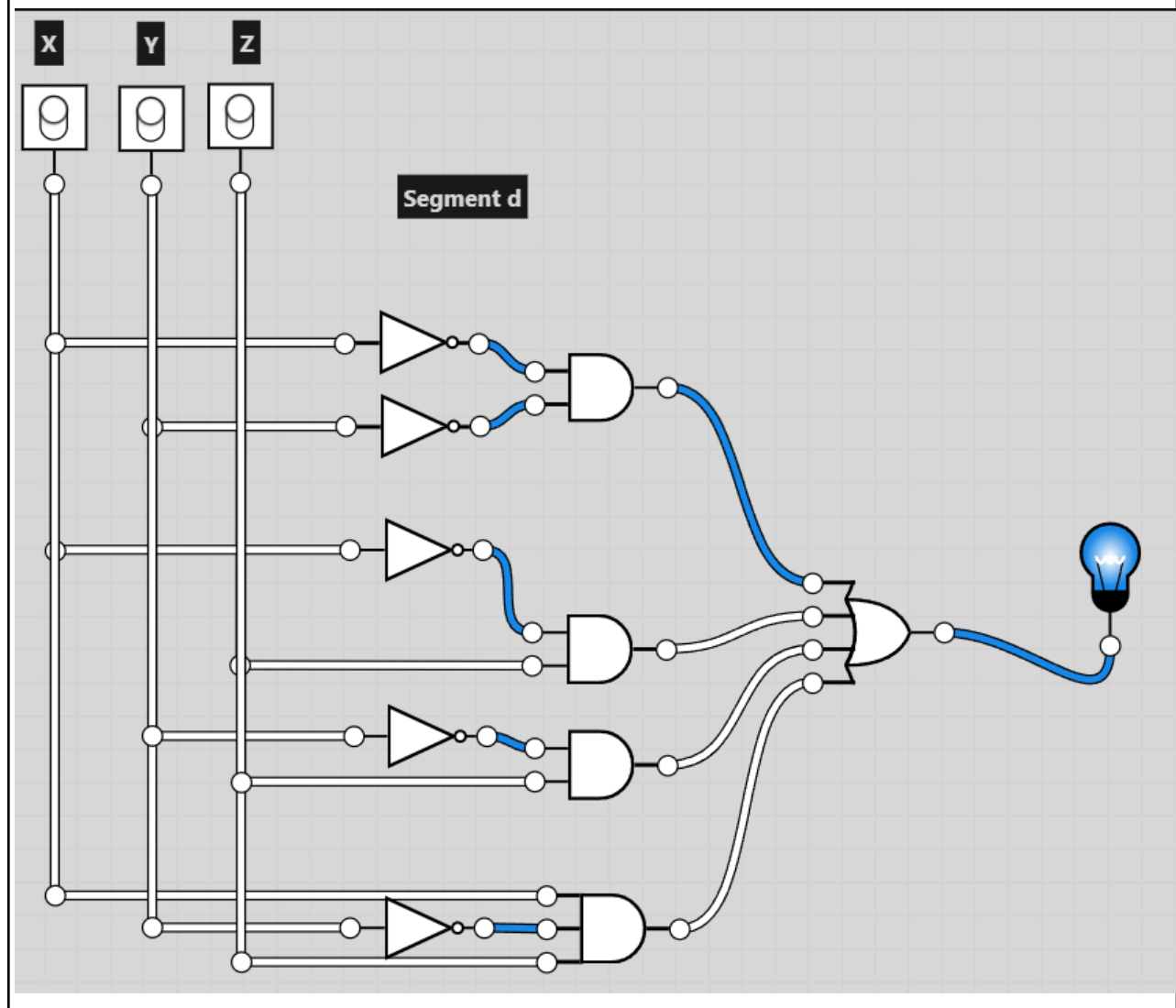
SEGMENT d:

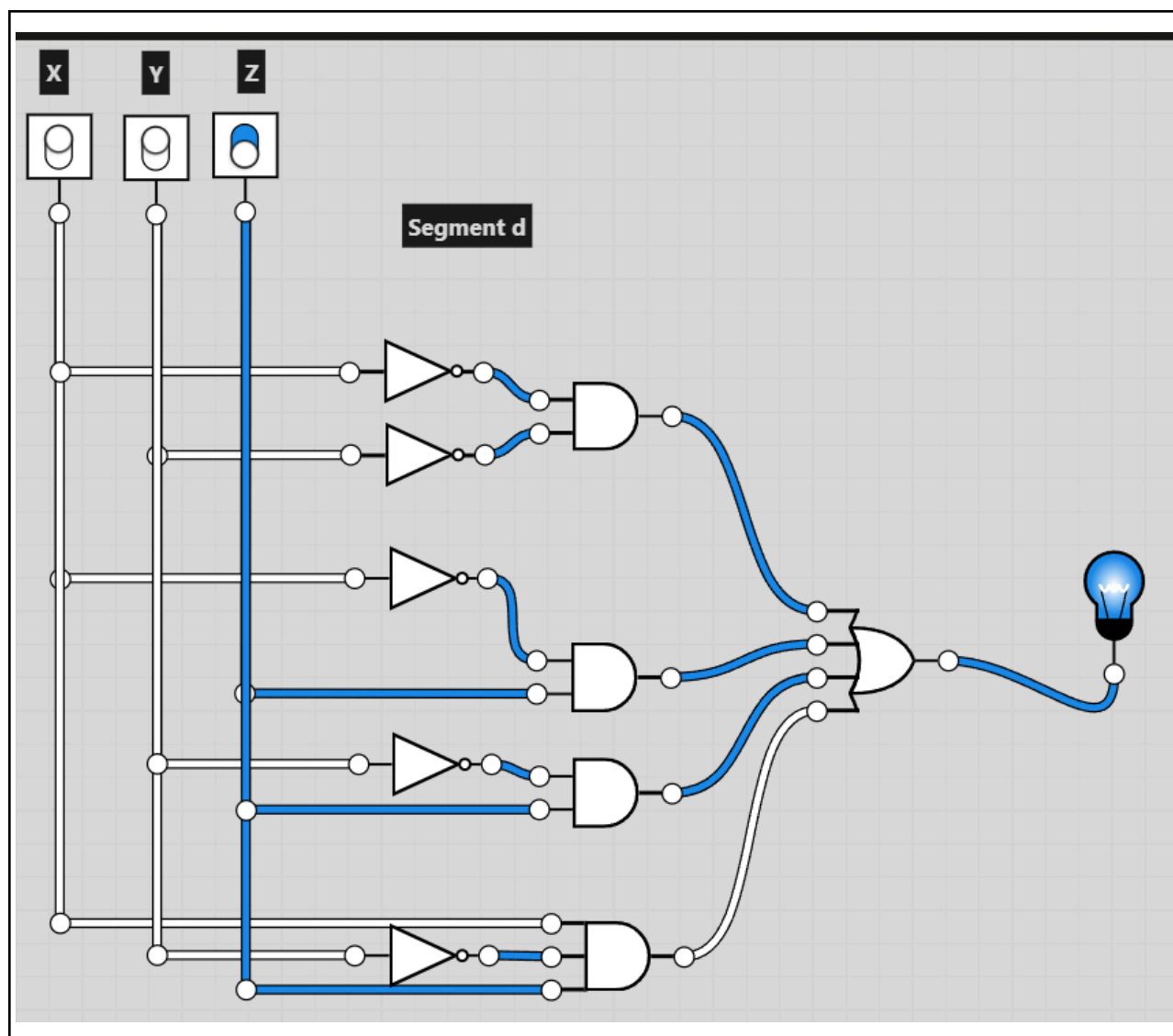
K-map:

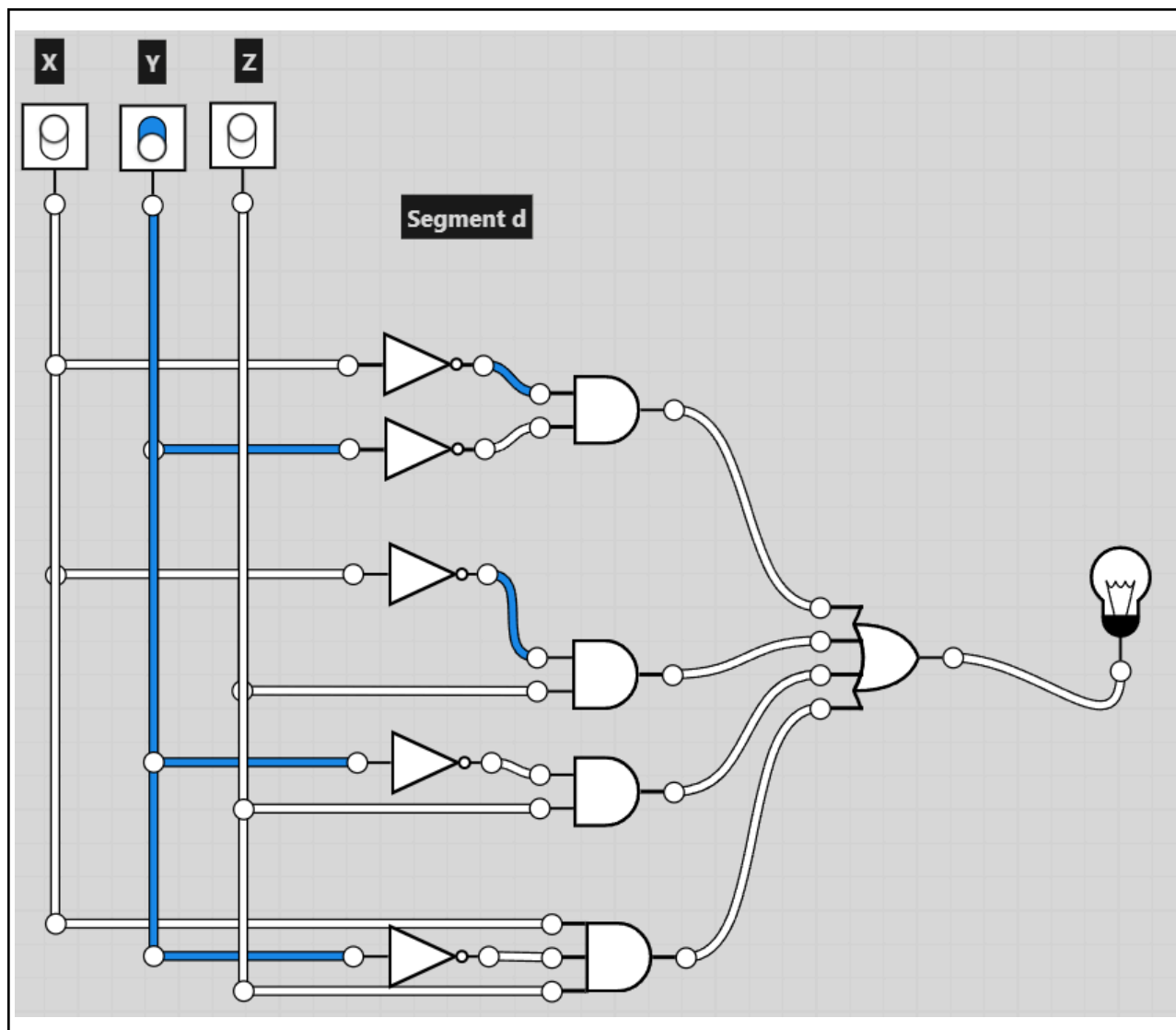
X \ YZ	00	01	11	10
0	1	1	1	0

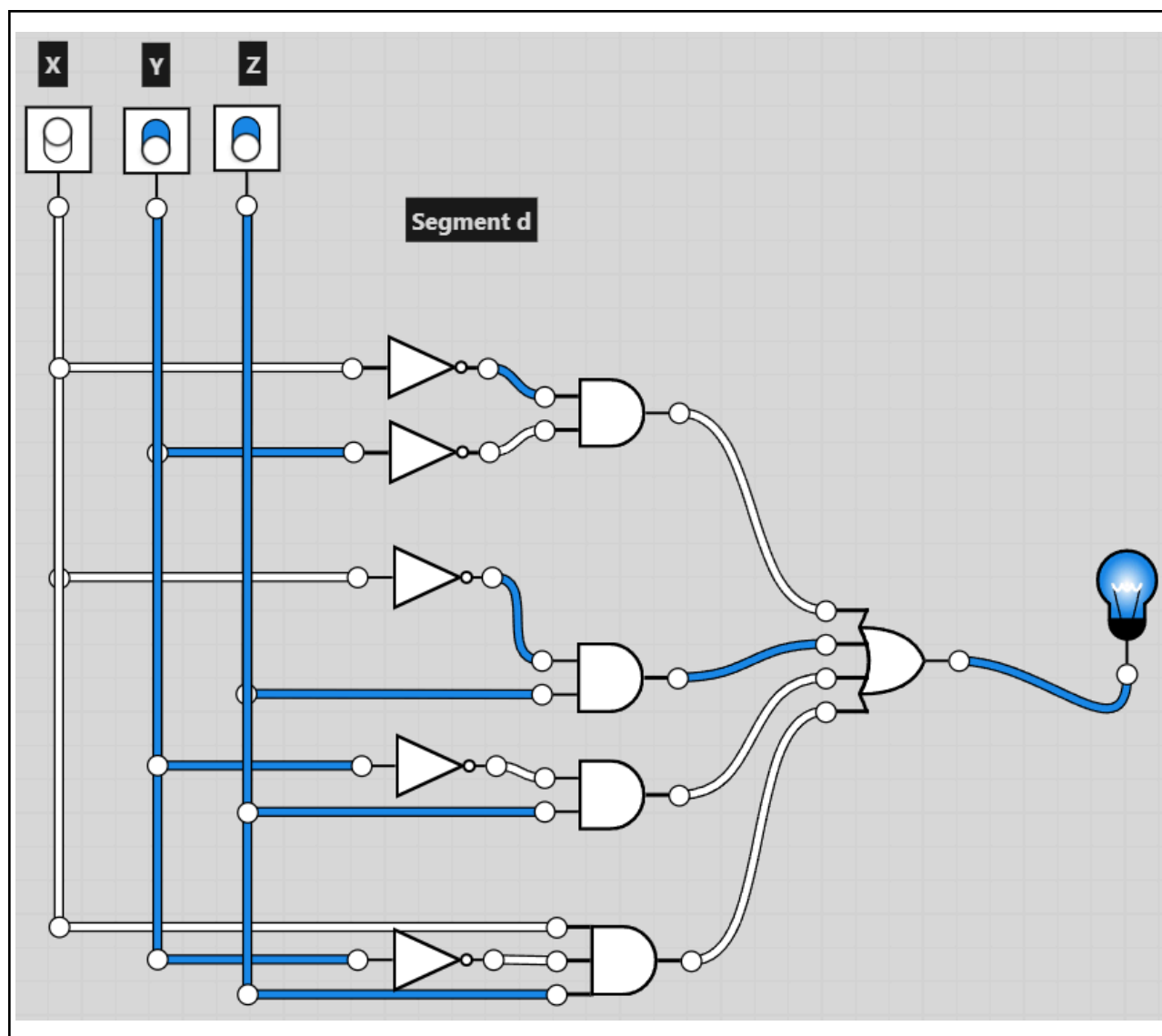
1	0	1	0	1
---	---	---	---	---

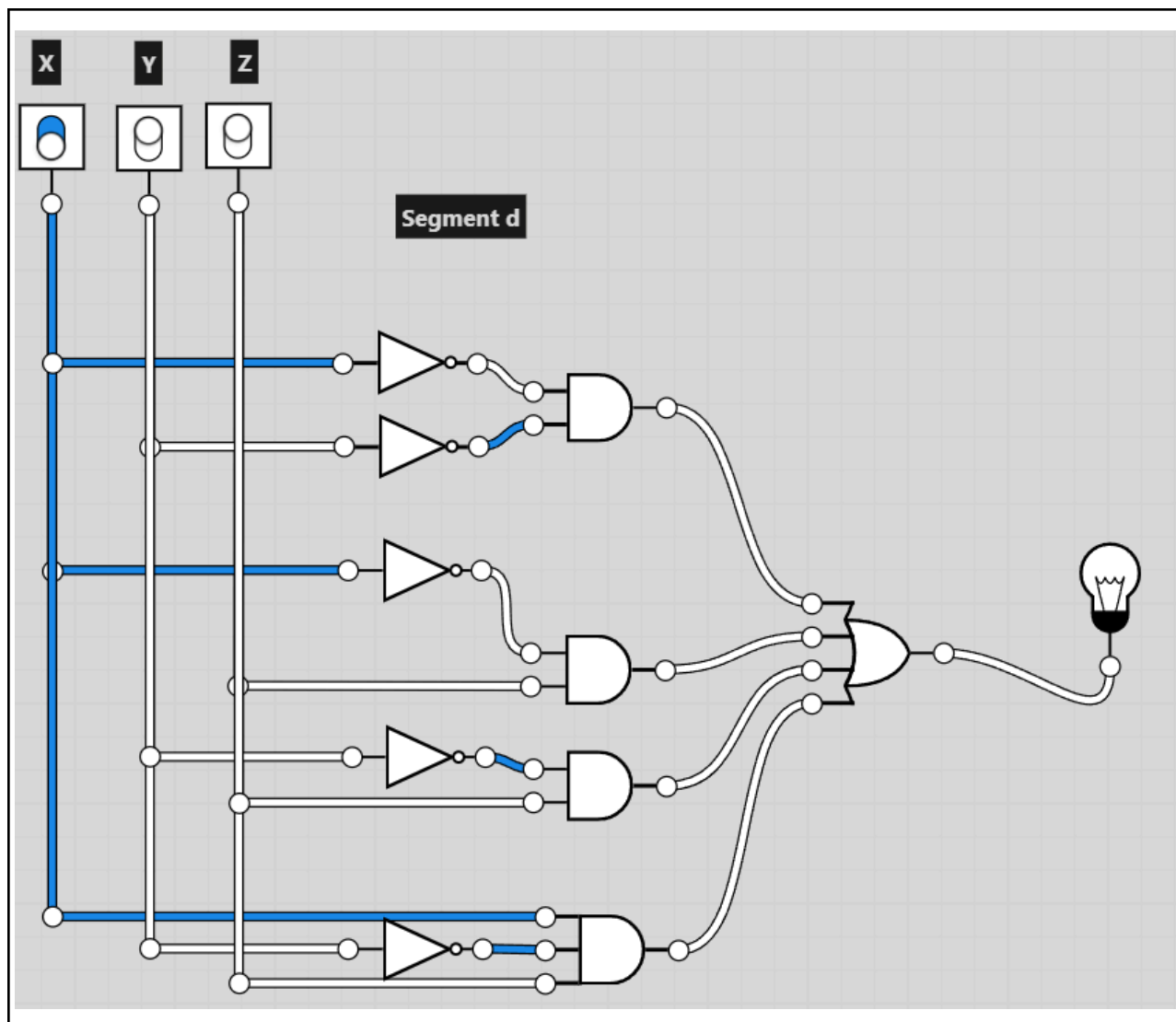
Logically:

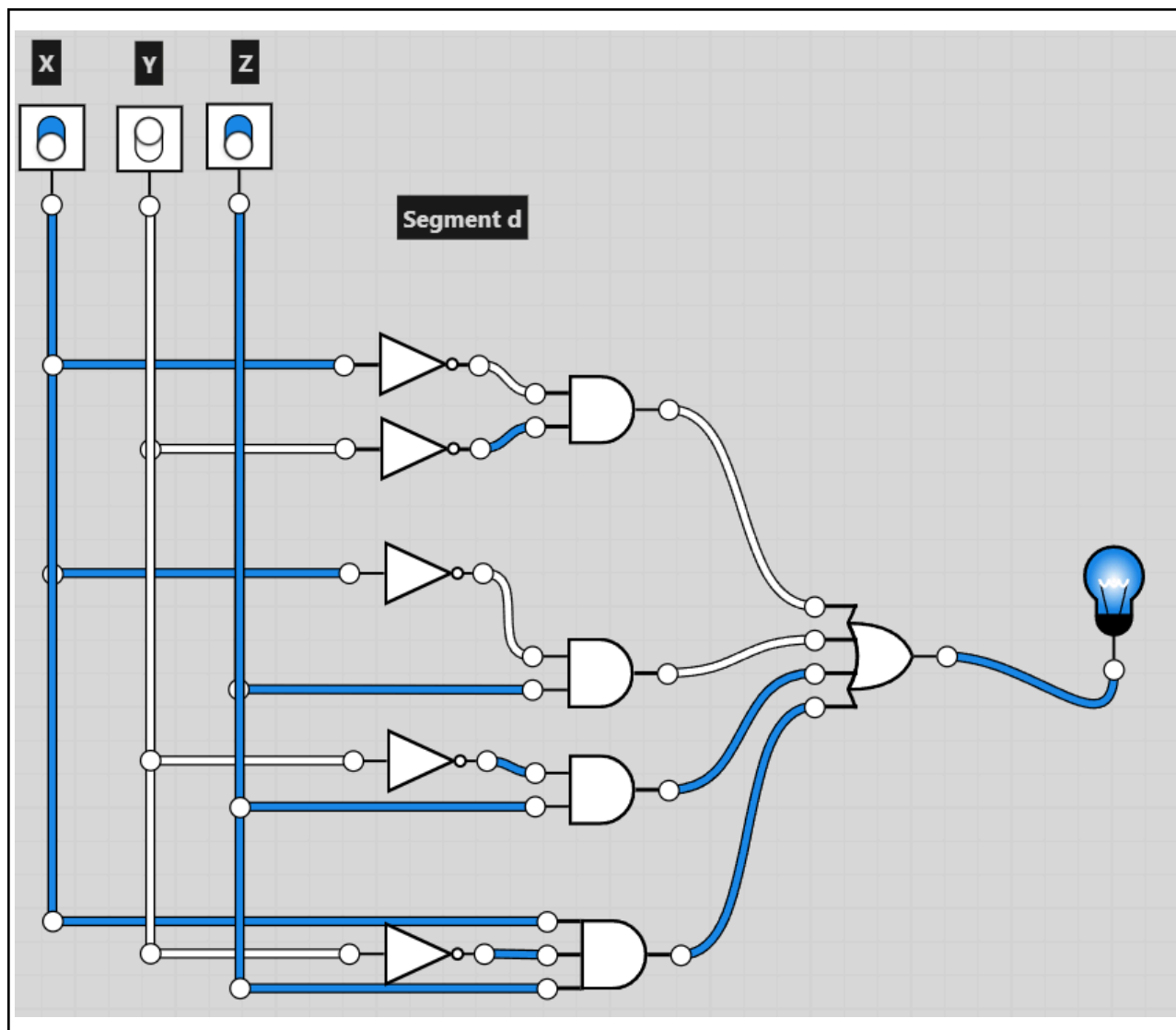


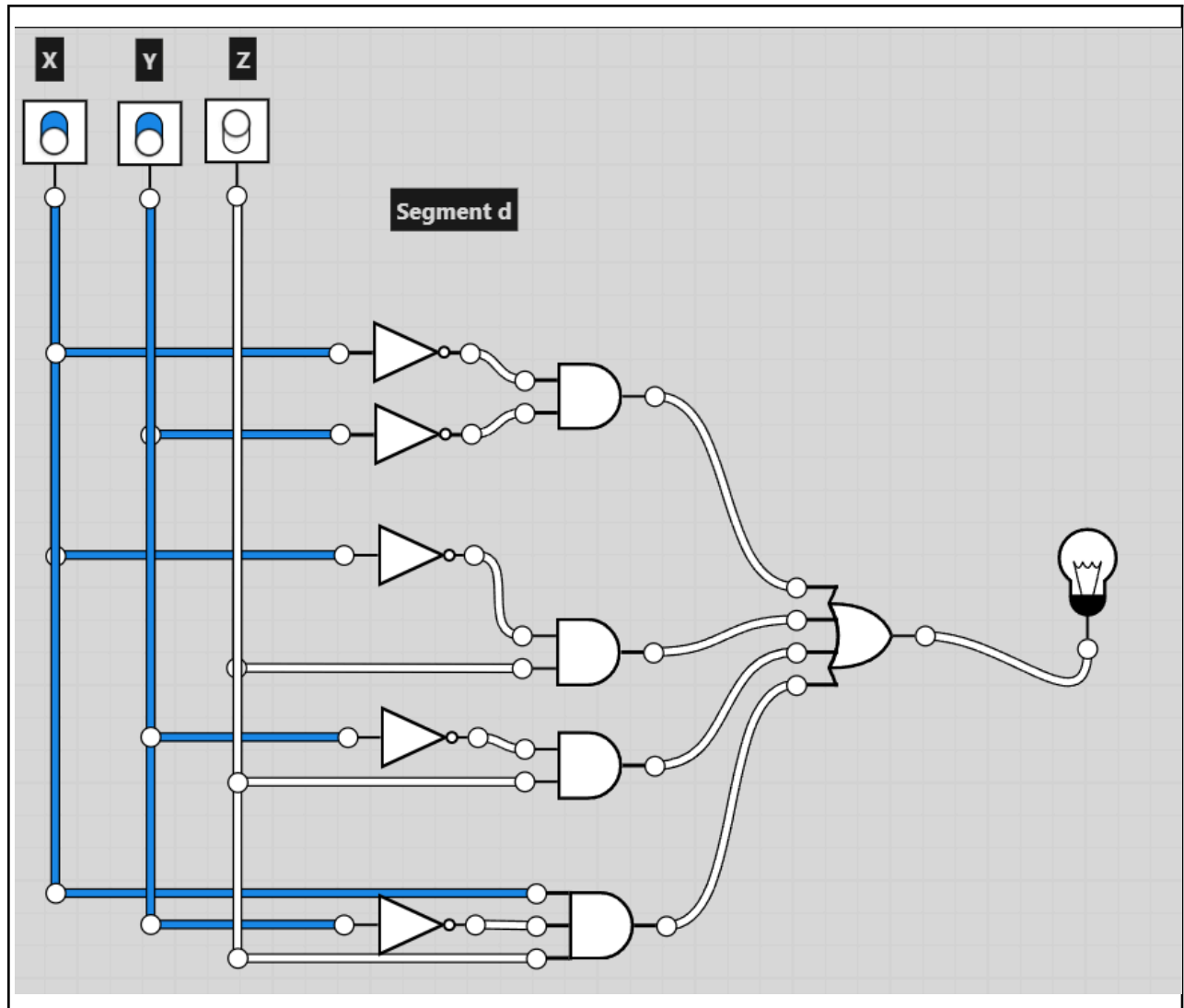


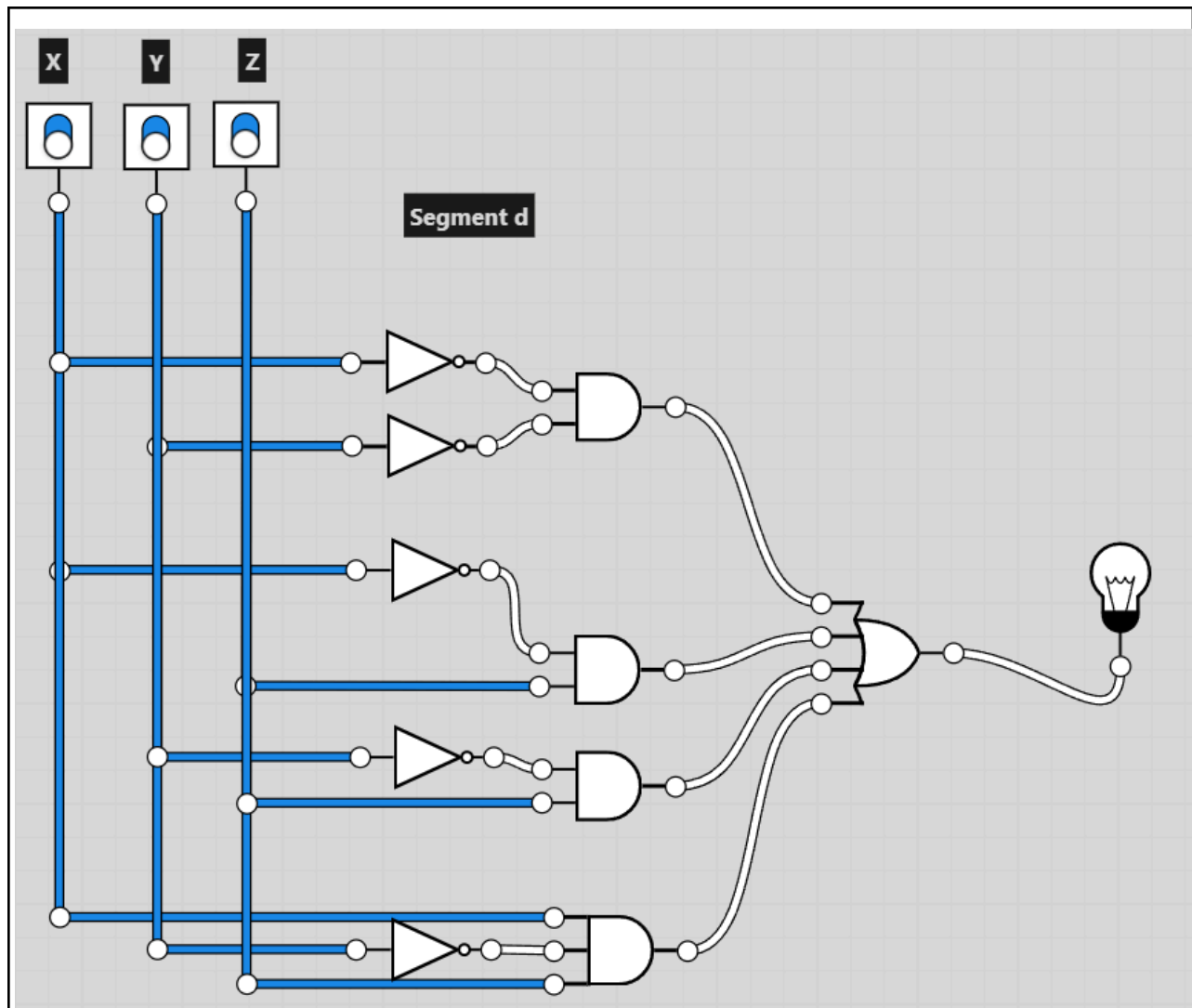












Boolean expression: $d = X'Y' + X'Z + Y'Z + XYZ$

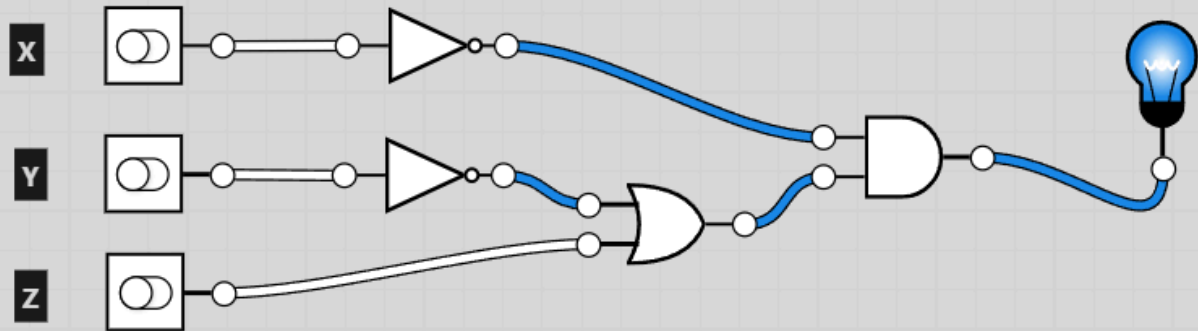
SEGMENT e:

K-map:

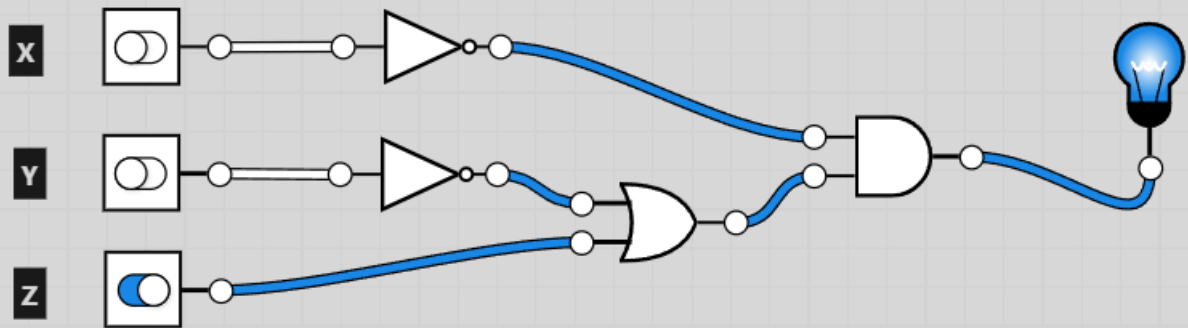
X \ YZ	00	01	11	10
0	1	1	1	0
1	0	0	0	0

Logically:

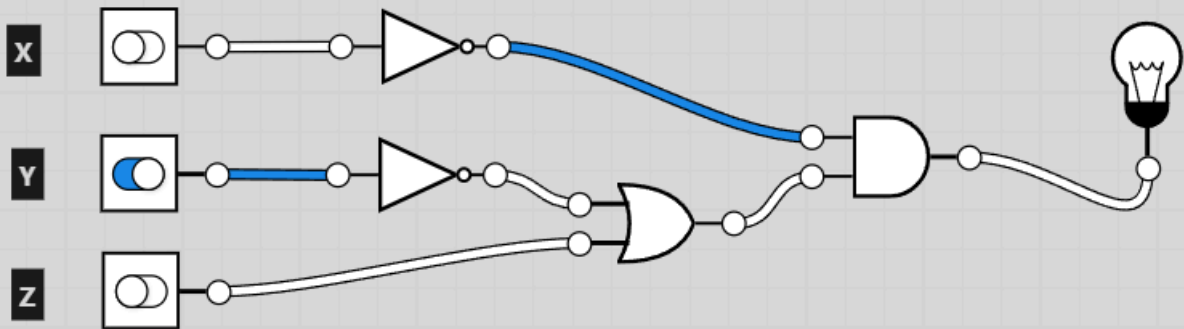
Segment e



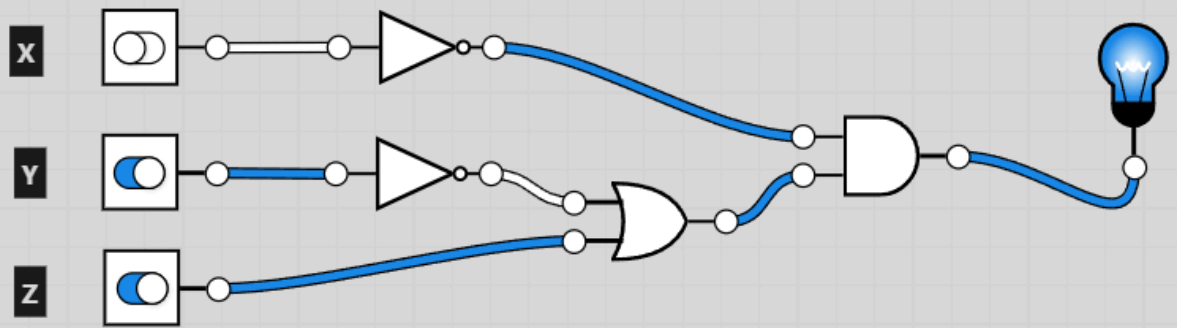
Segment e



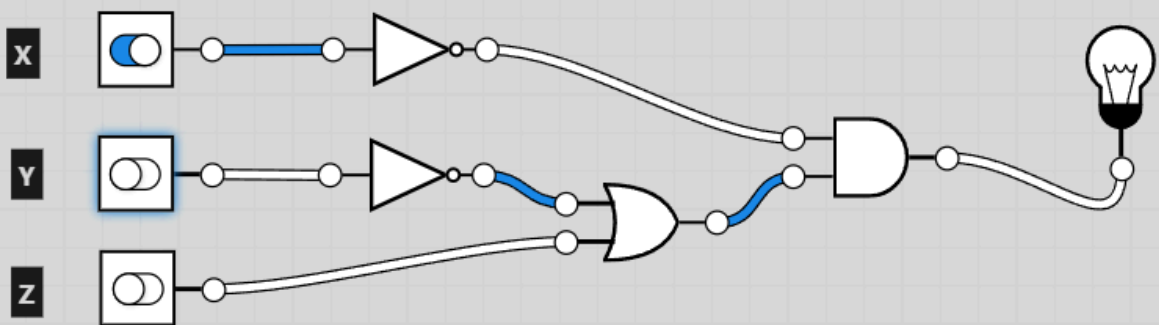
Segment e



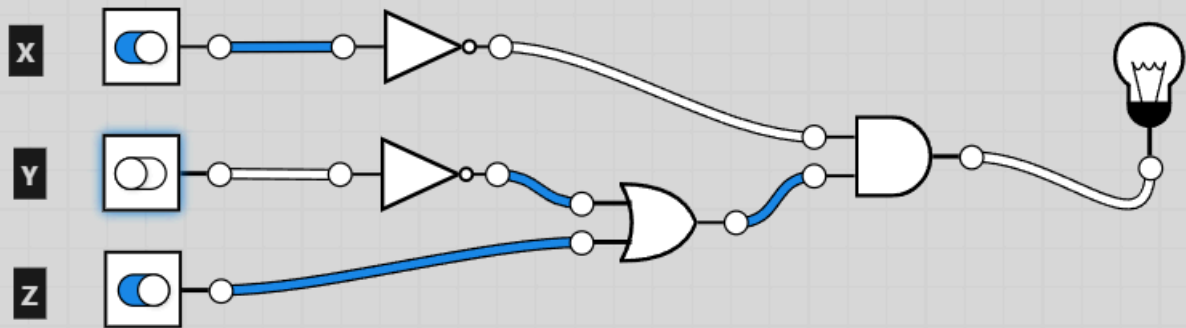
Segment e



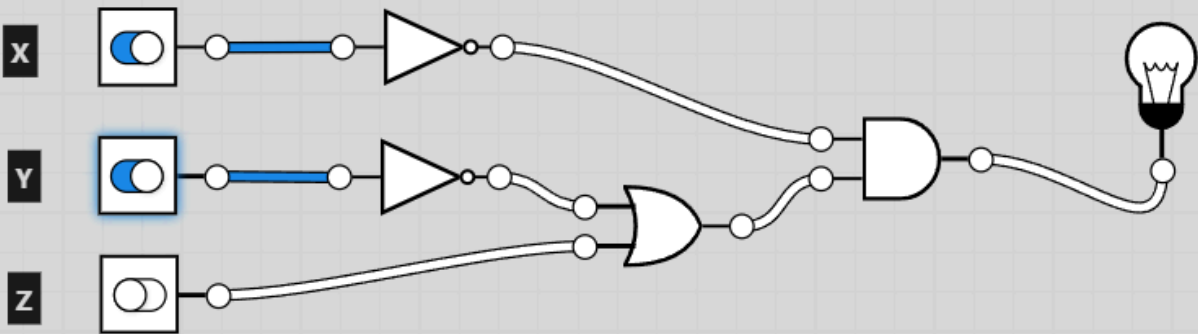
Segment e



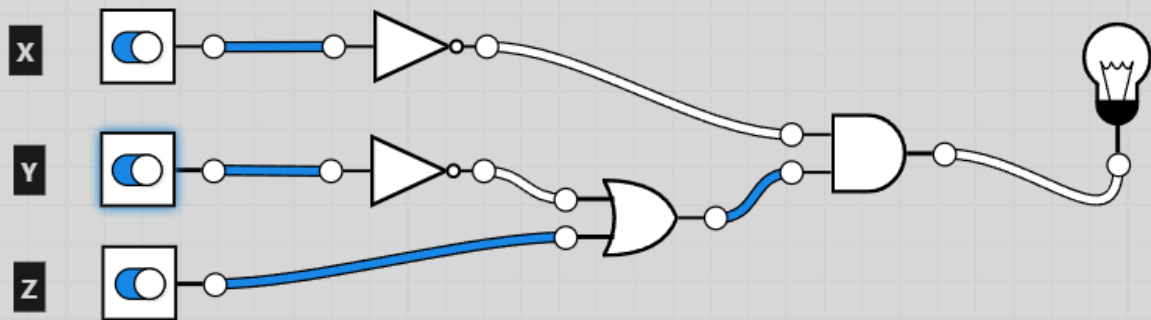
Segment e



Segment e



Segment e



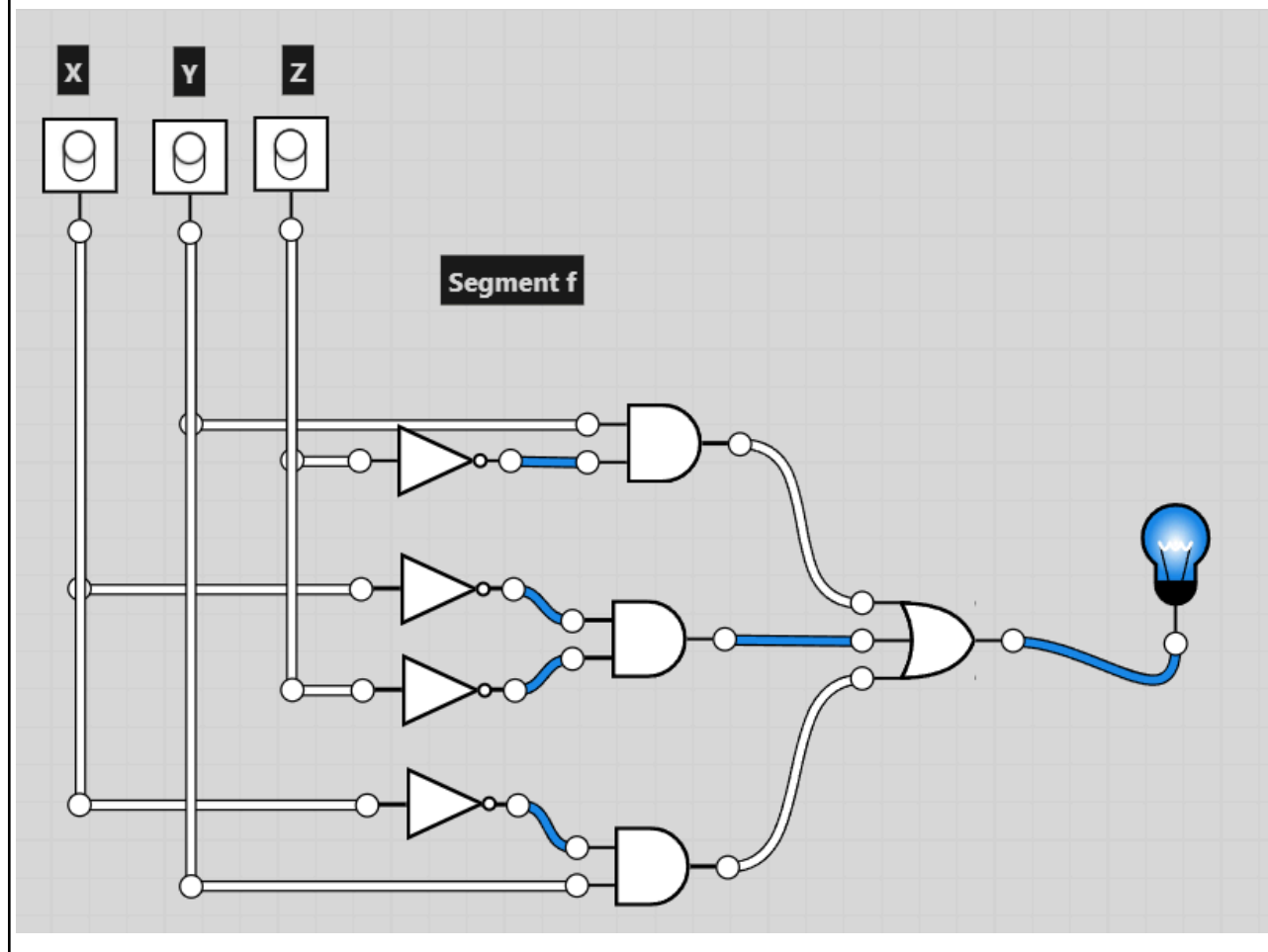
Boolean expression: $e = X'(Z+Y')$

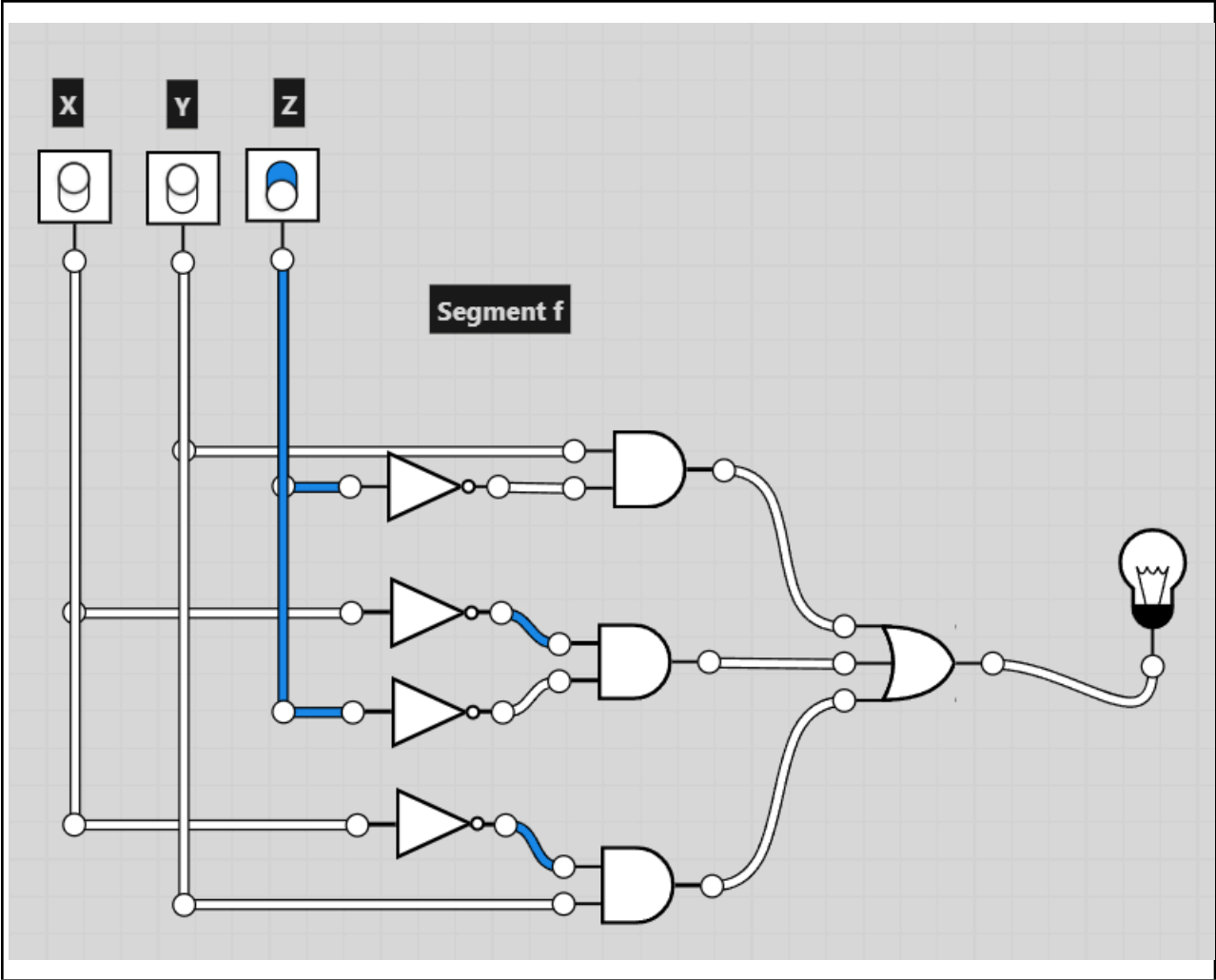
SEGMENT f:

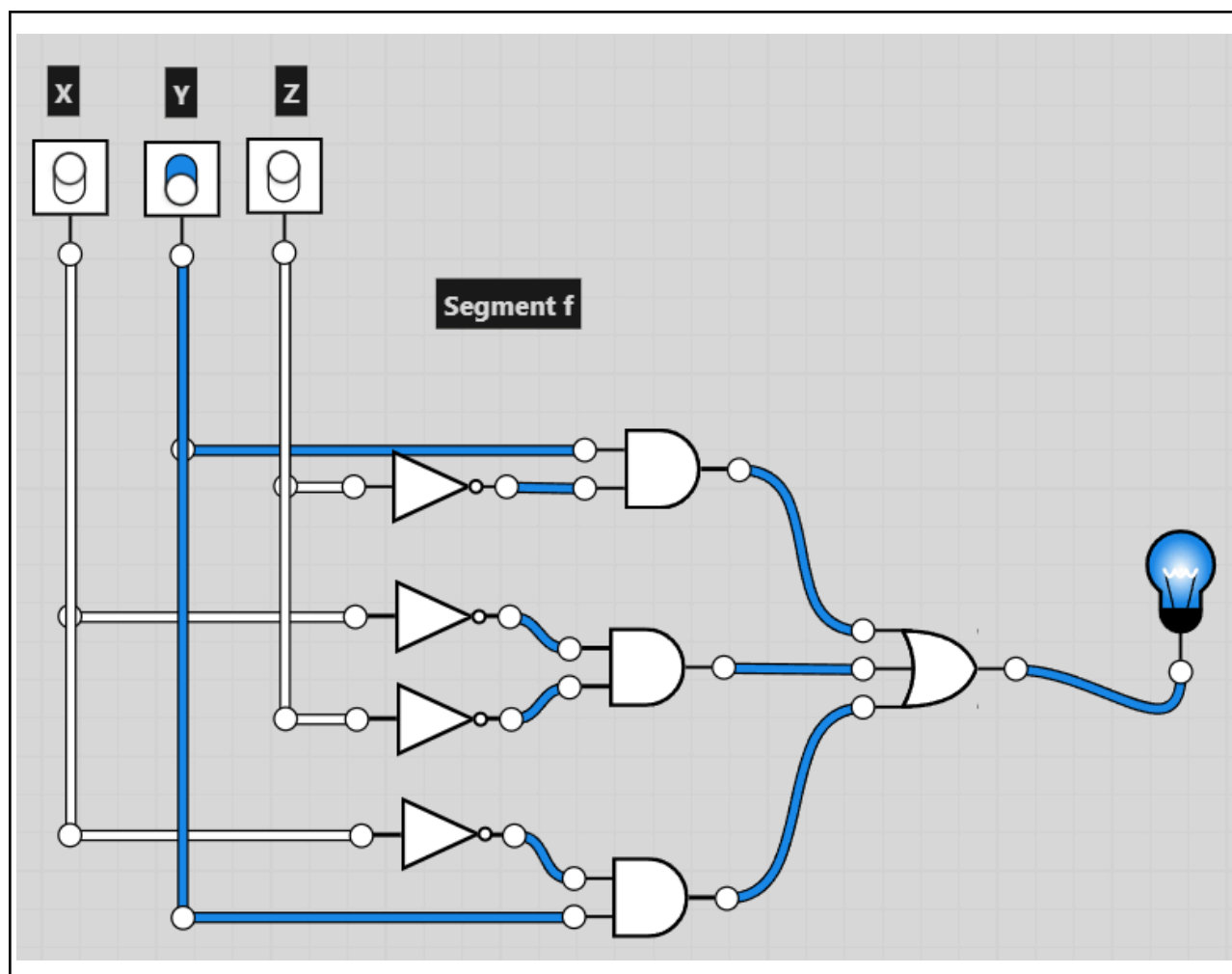
K-map:

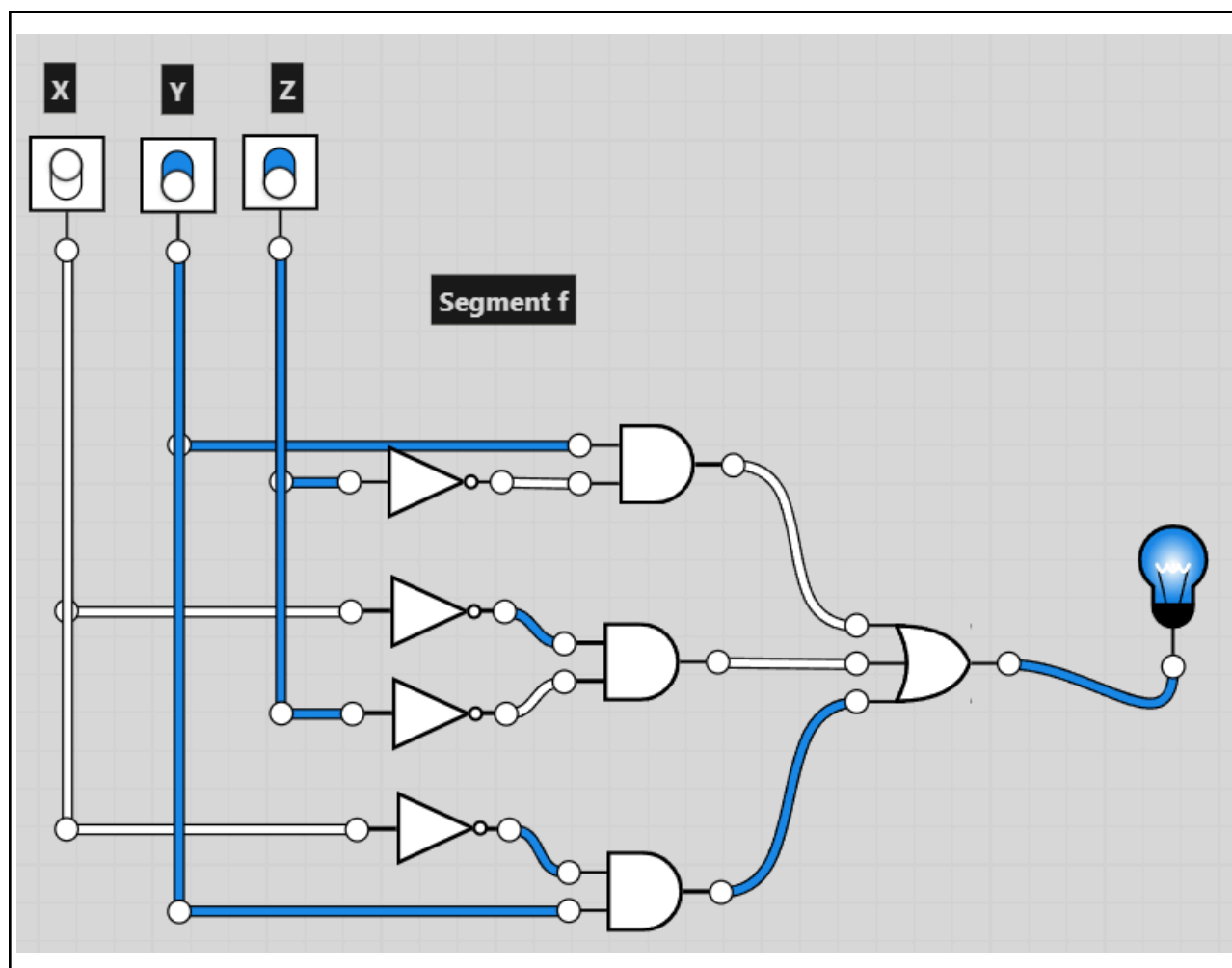
X \ YZ	00	01	11	10
0	1	0	1	1
1	0	0	0	1

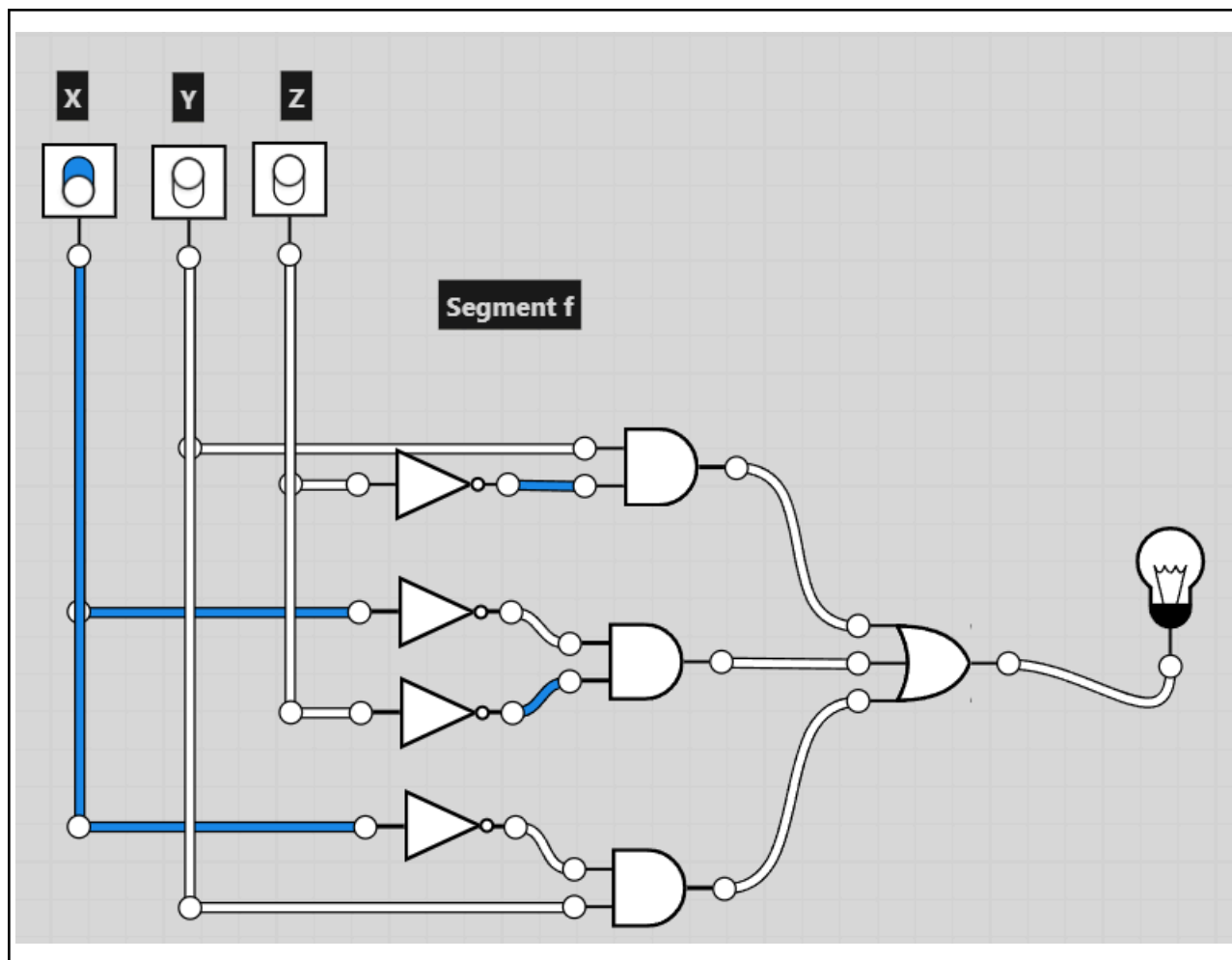
Logicly:

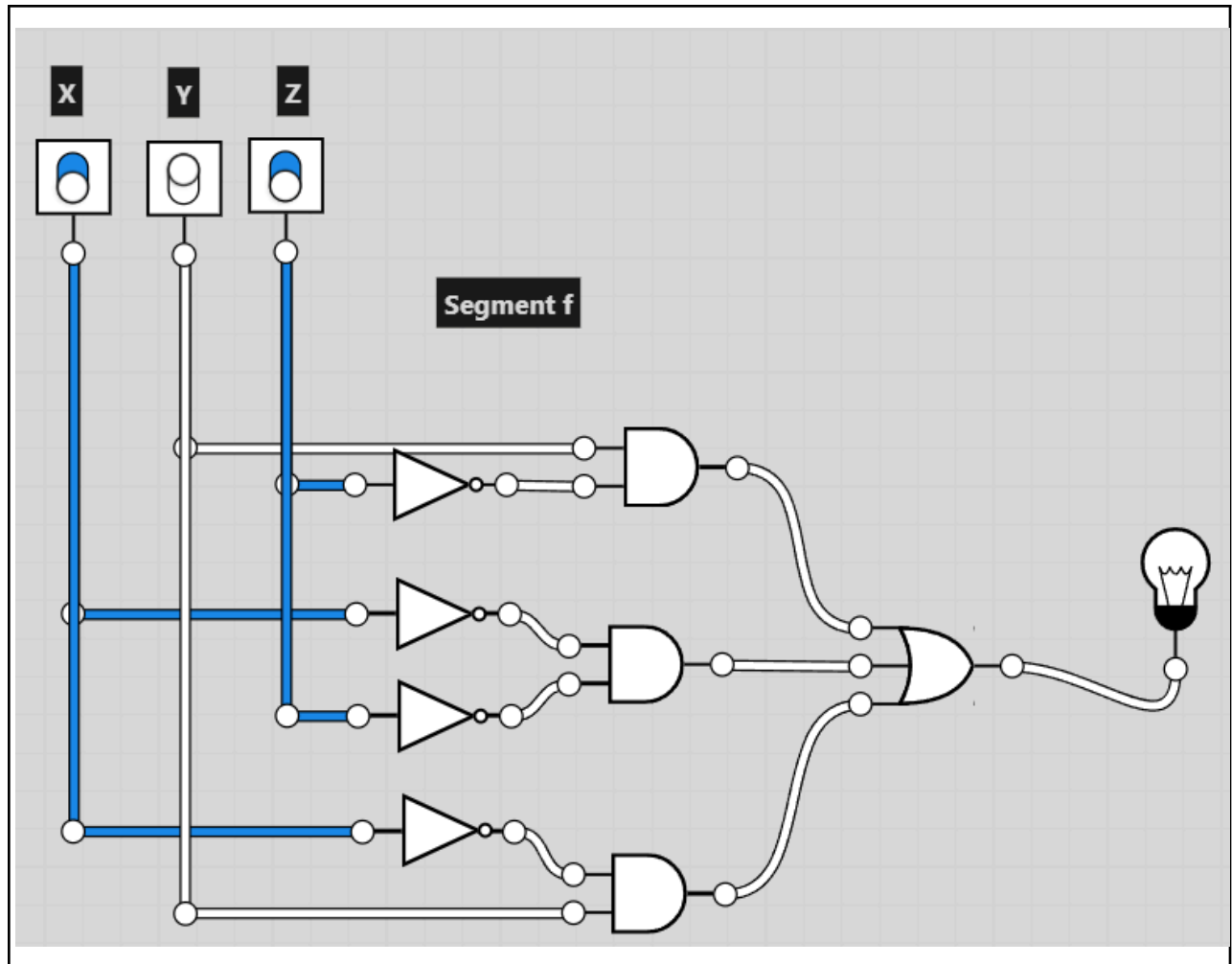


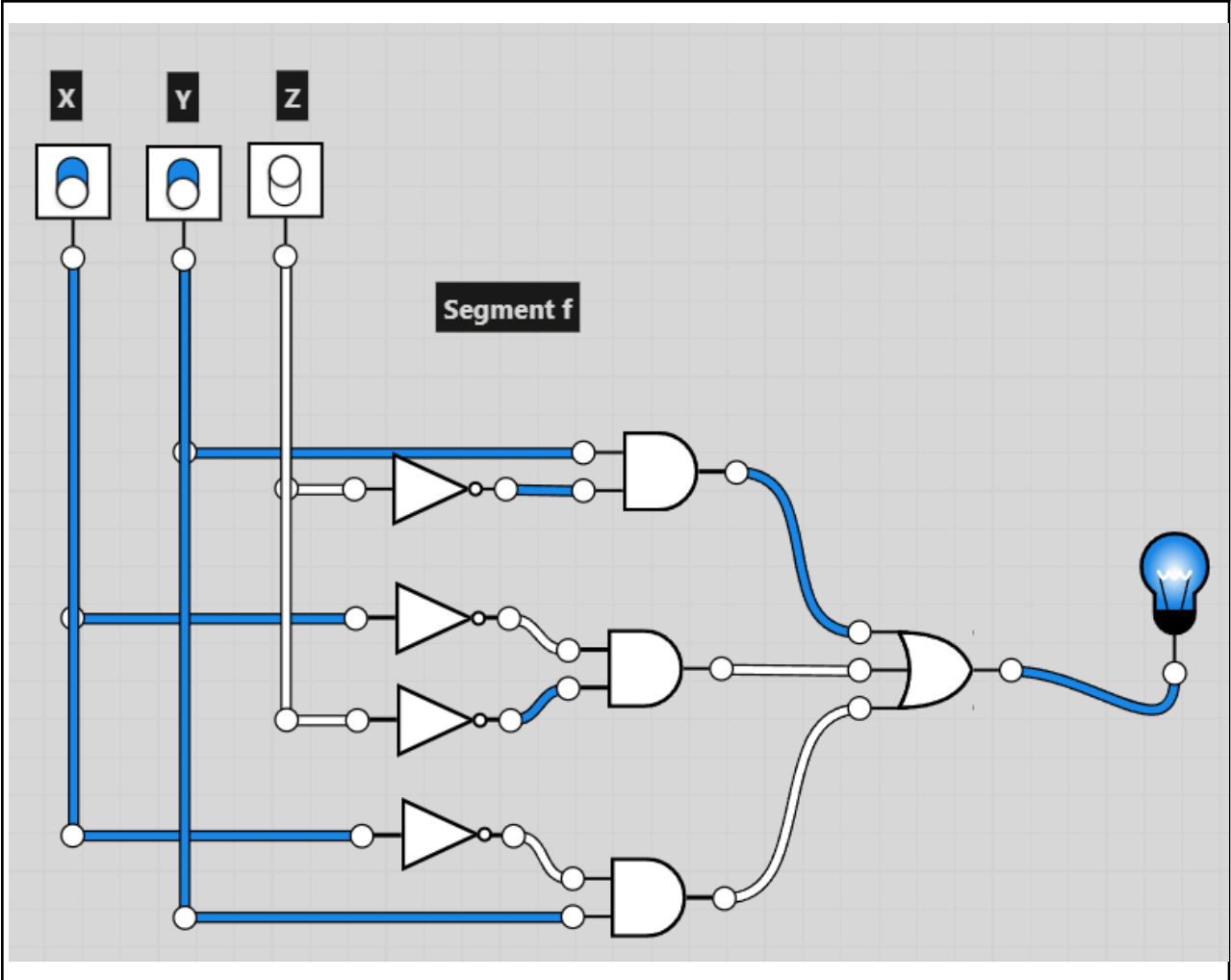


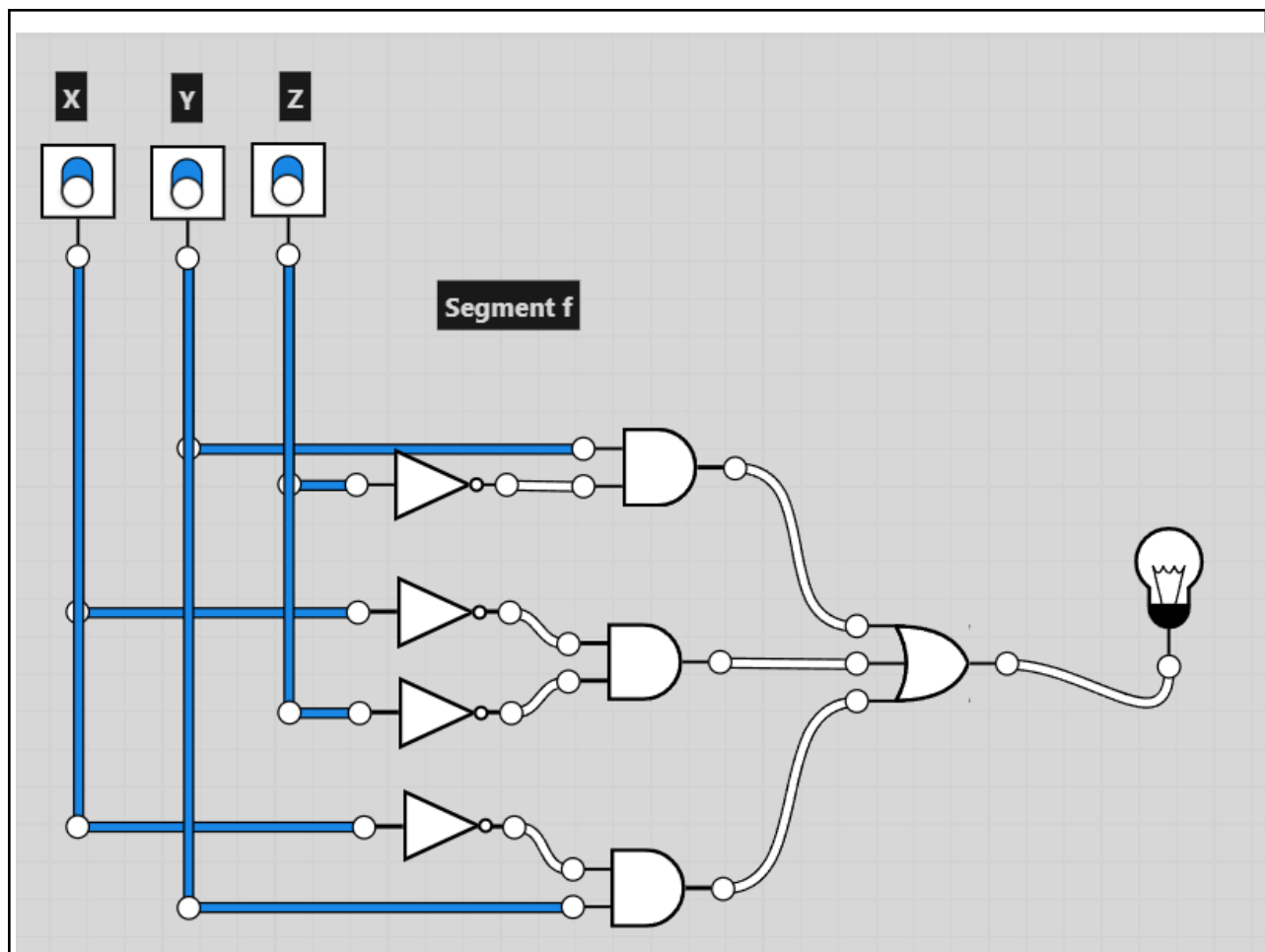












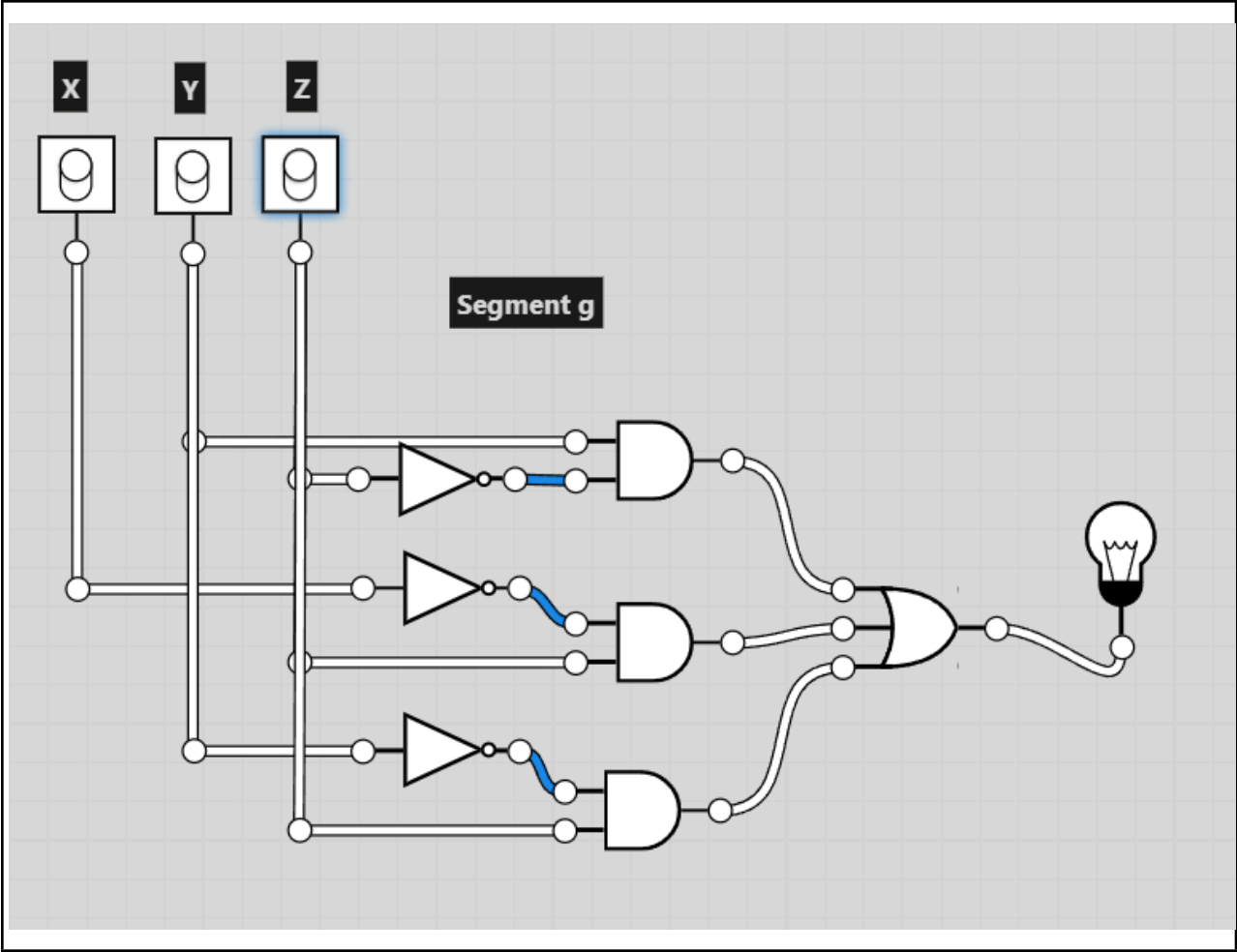
Boolean expression: $f = YZ' + X'Z + XY$

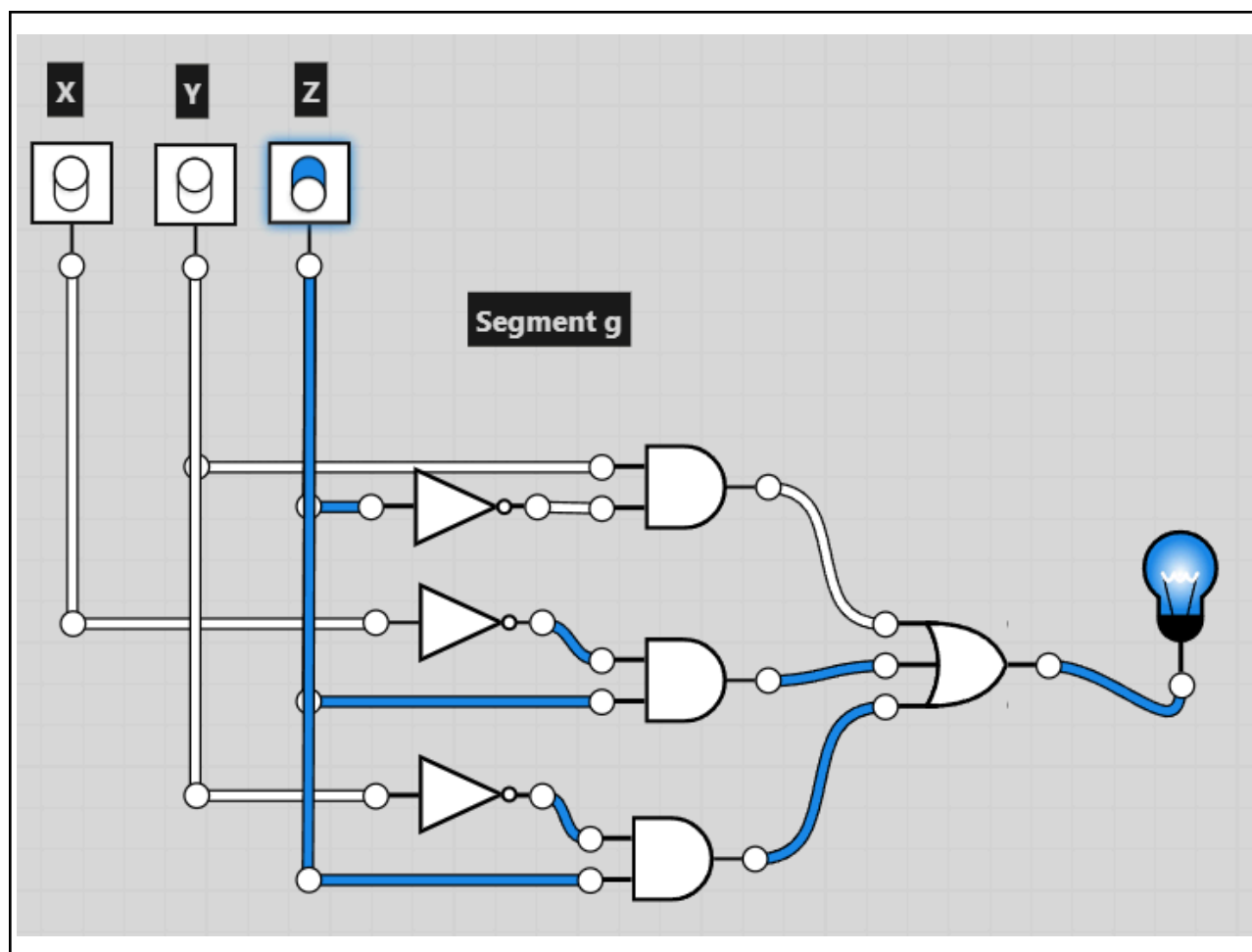
SEGMENT g:

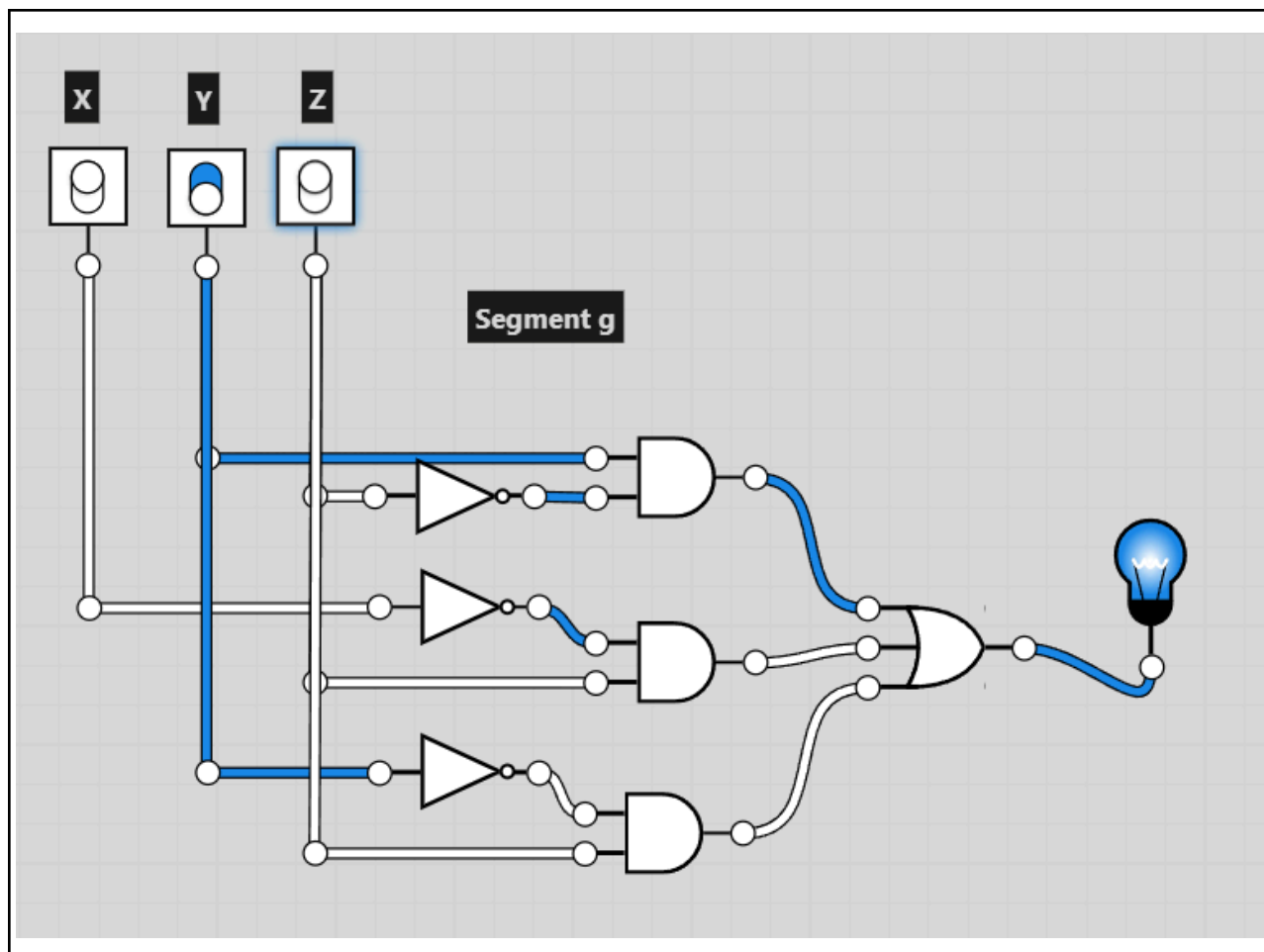
K-map:

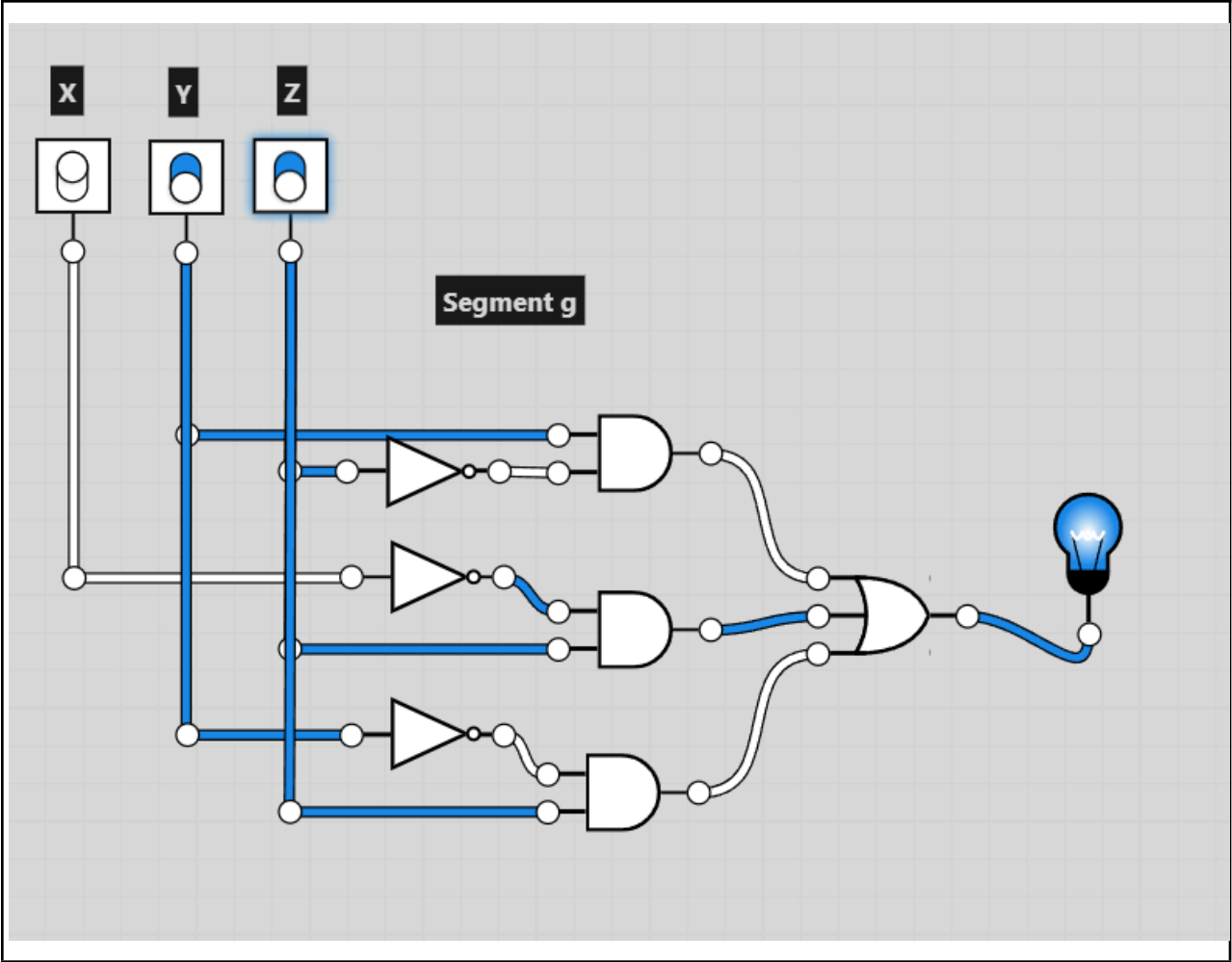
X \ YZ	00	01	11	10
0	0	1	1	1
1	0	1	0	1

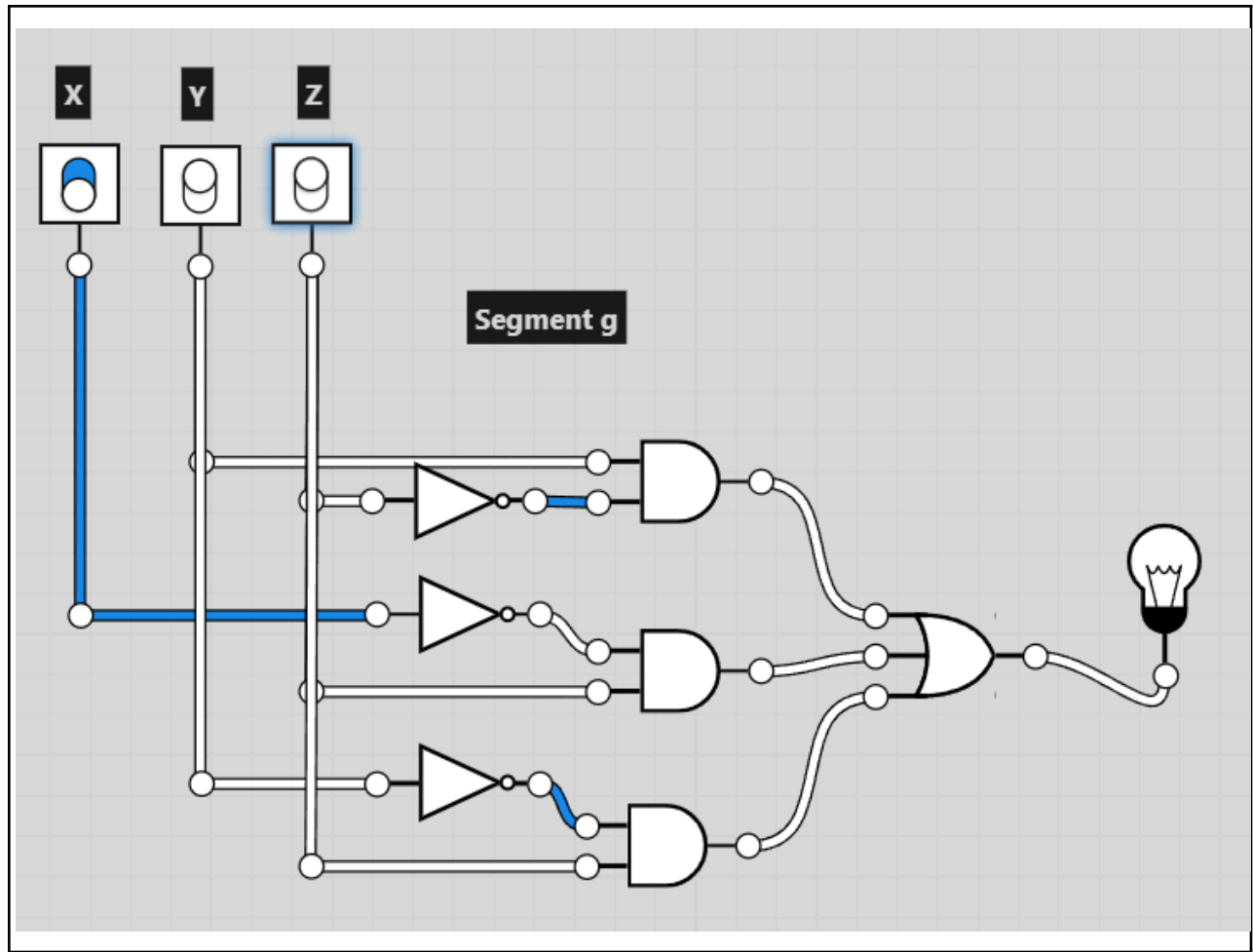
Logically:

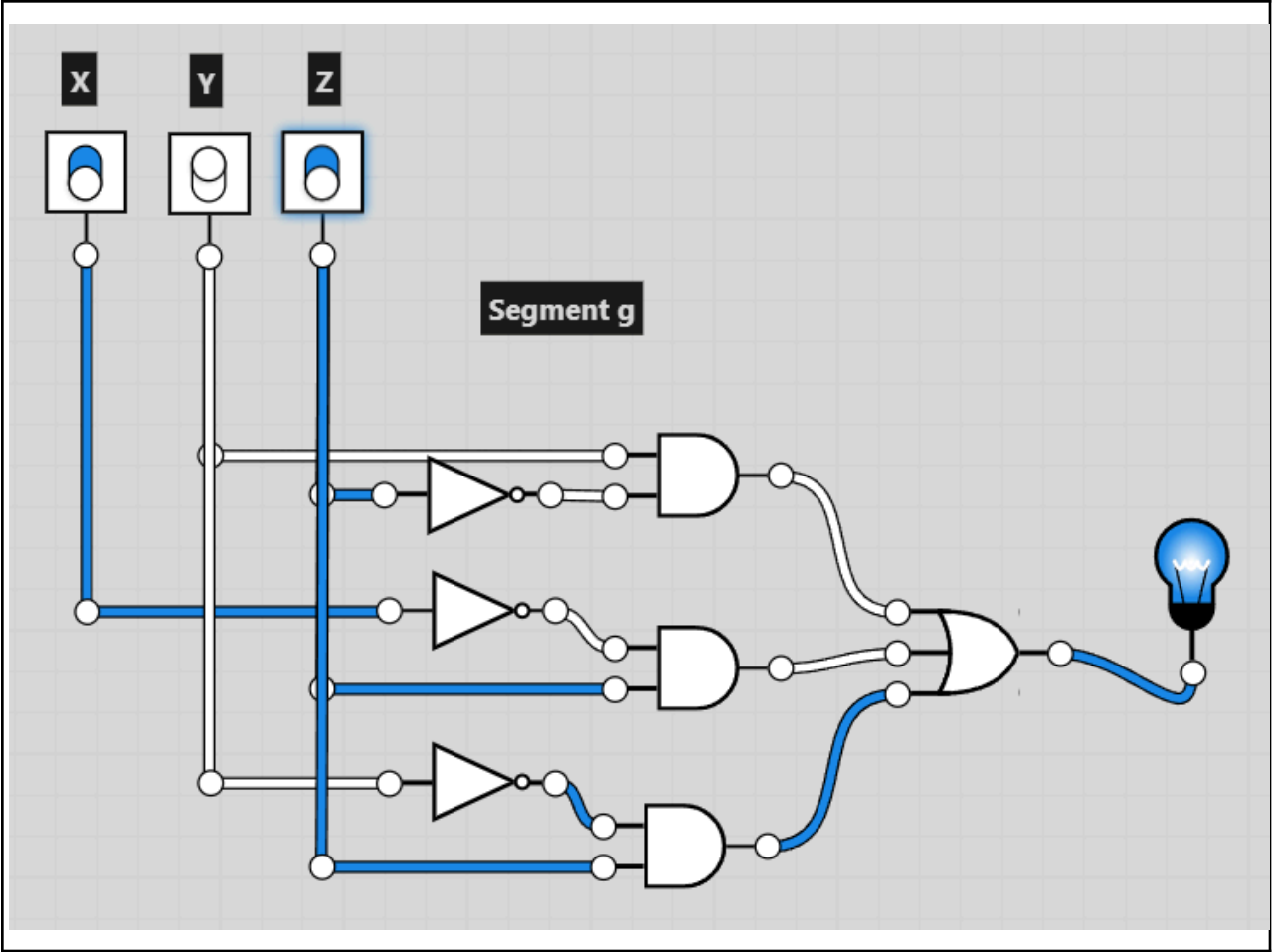


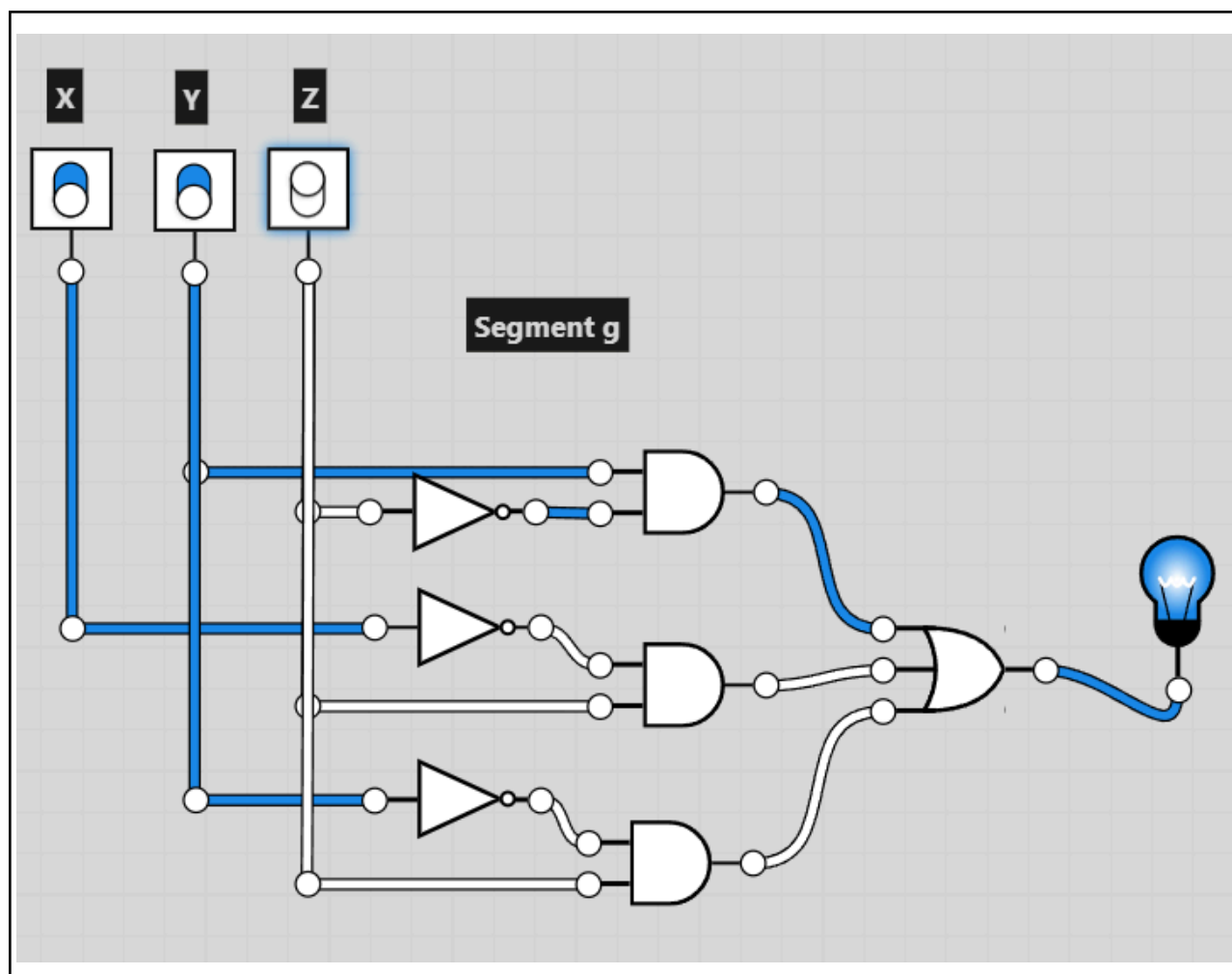


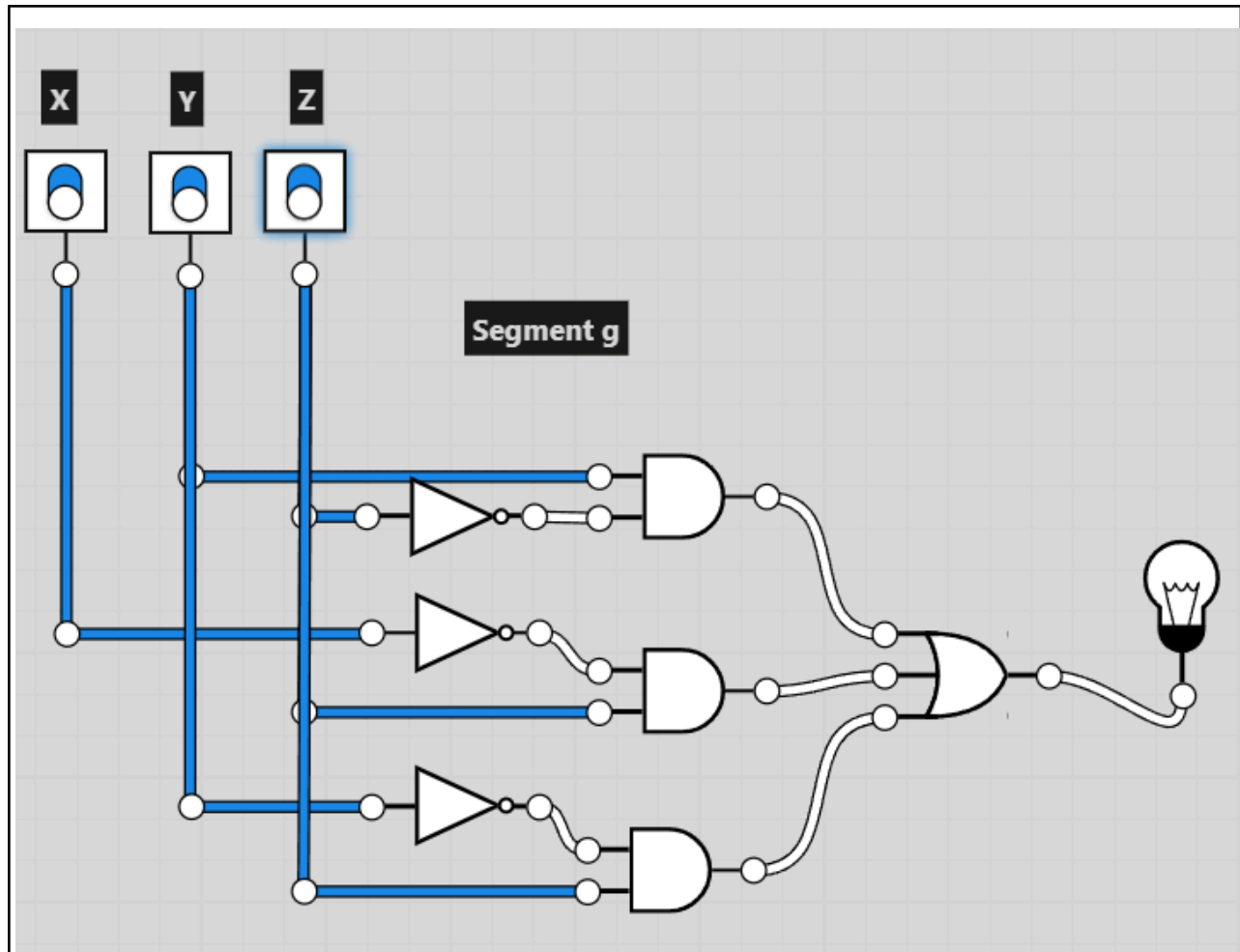










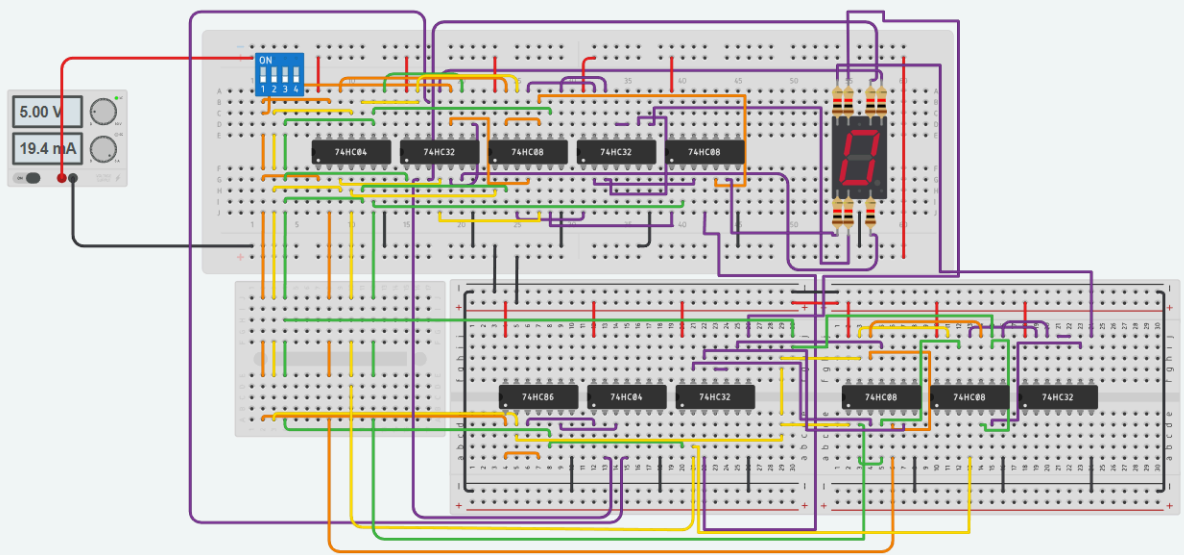


Boolean expression: $g = YZ' + X'Z + Y'Z$

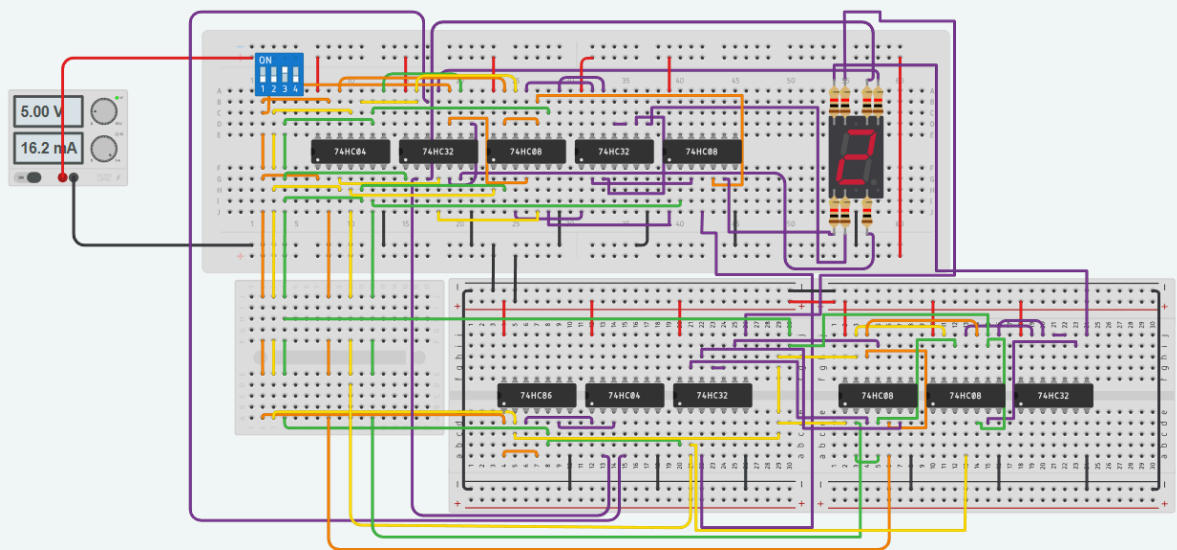
6. Final Output:

Screenshot or print screen of TinkercAD application different output simulations (5 points deductions for only one output in every required circuit). Also, upload a video of the simulated circuit naming or talking about the shown input combinations. Failure to comply to each instruction will result to 5 points deductions. Please include the recorded simulation video in this section and write at least one sentence definition or label for each figure or image.

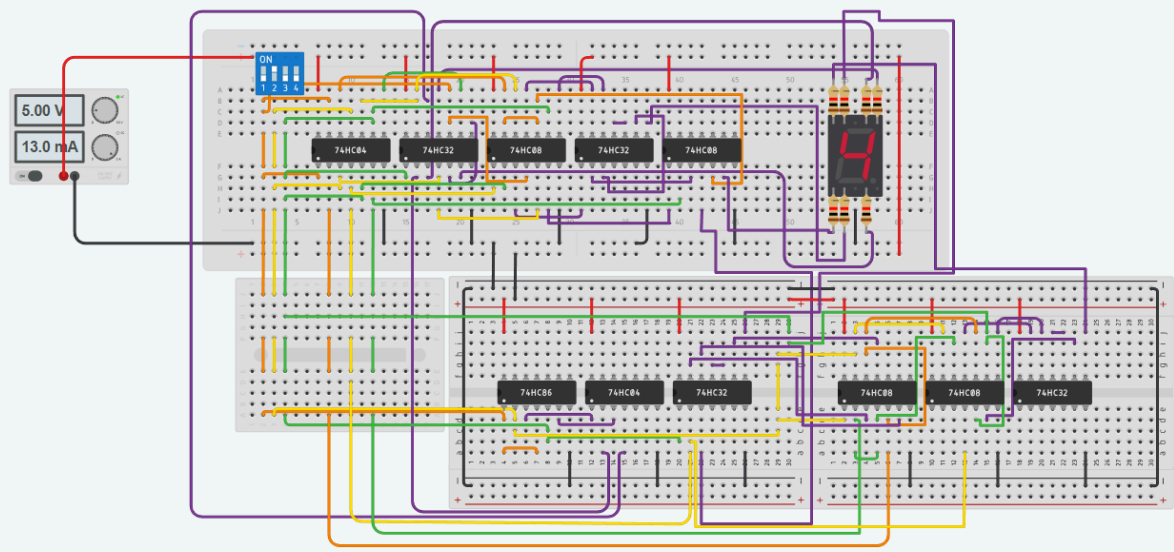
TINKERCAD SCREENSHOTS



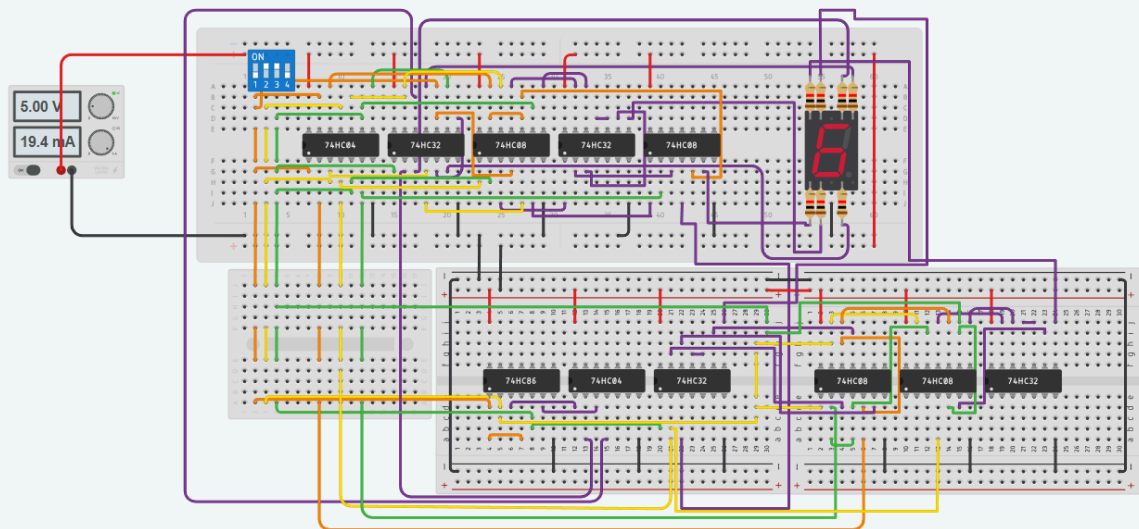
0-0-0



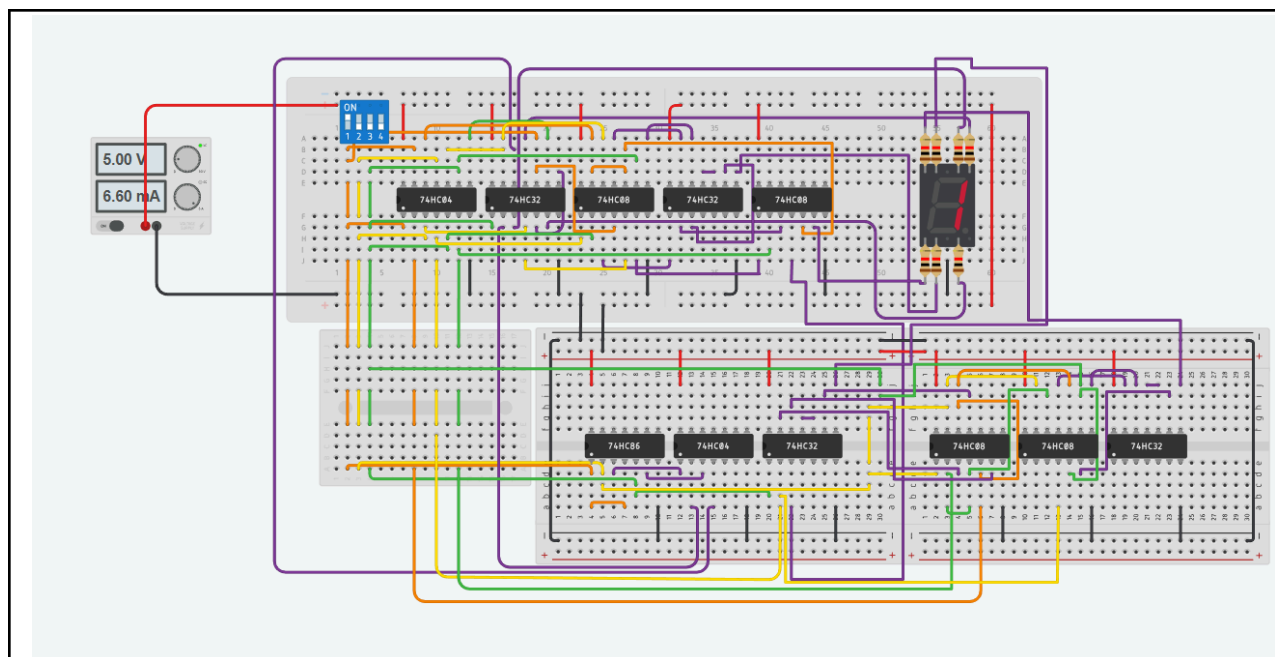
0-0-1



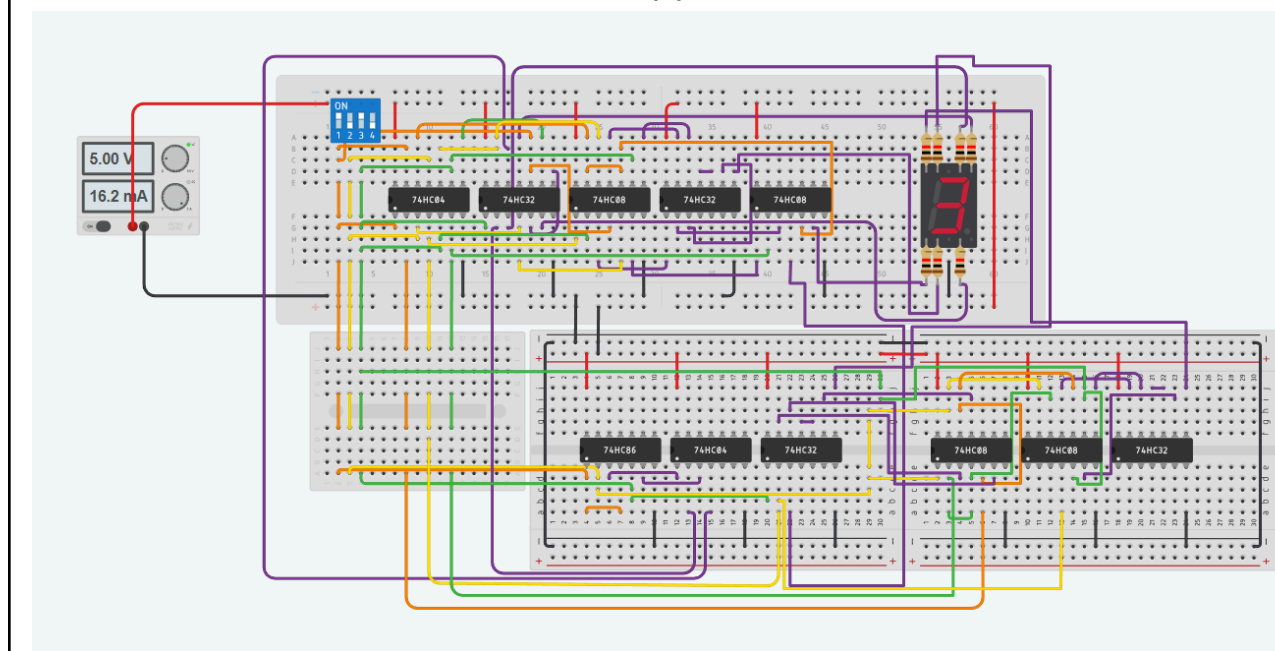
0-1-0



0-1-1



1-0-0



1-0-1

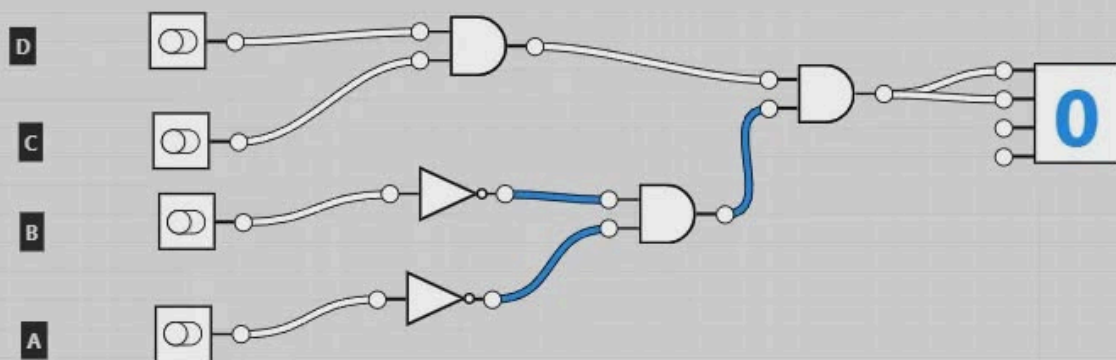
and we create an exact truth table for each potential input of odd and even numbers. We discovered the importance of the kmap technique for reducing the complicated boolean functions to their most basic sum of products form. In particular, we derived the logic statements for each of the seven segments (a through g) by using K-Map to generate the expected output generated in the TinkerCAD application. Our understanding of how to use a decoder combinational logic circuit with the Seven Segment display to show binary data was strengthened during the entire experiment. A physical relationship between logical logic and useful electronics was made possible by the physical implementation, which was first simulated in Logically and later in TinkerCAD. We first came through a problem with the circuit displaying the number "6" incorrectly in the even sequence during the activity. We discovered that the minterms for segment 'f' were incorrectly grouped on the K-Map, leading to an inaccurate logic expression. We learned from this mistake how important it is to thoroughly double-check each stage of the design process, especially the final wiring connections and the simplification of the K-Map. After fixing this error, we were able to complete a functional, effective circuit, showcasing both effective design and crucial troubleshooting abilities.

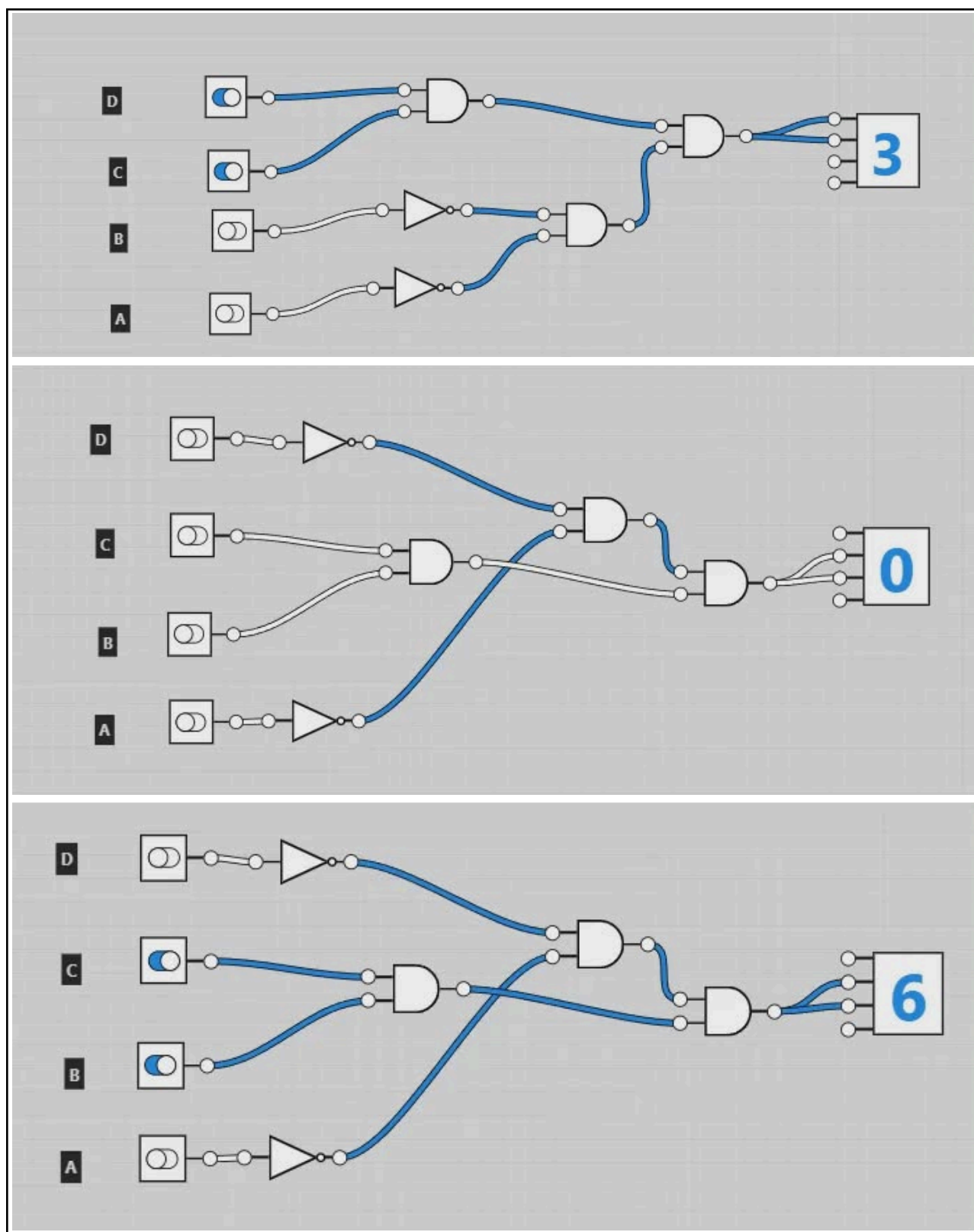
8. Questions:

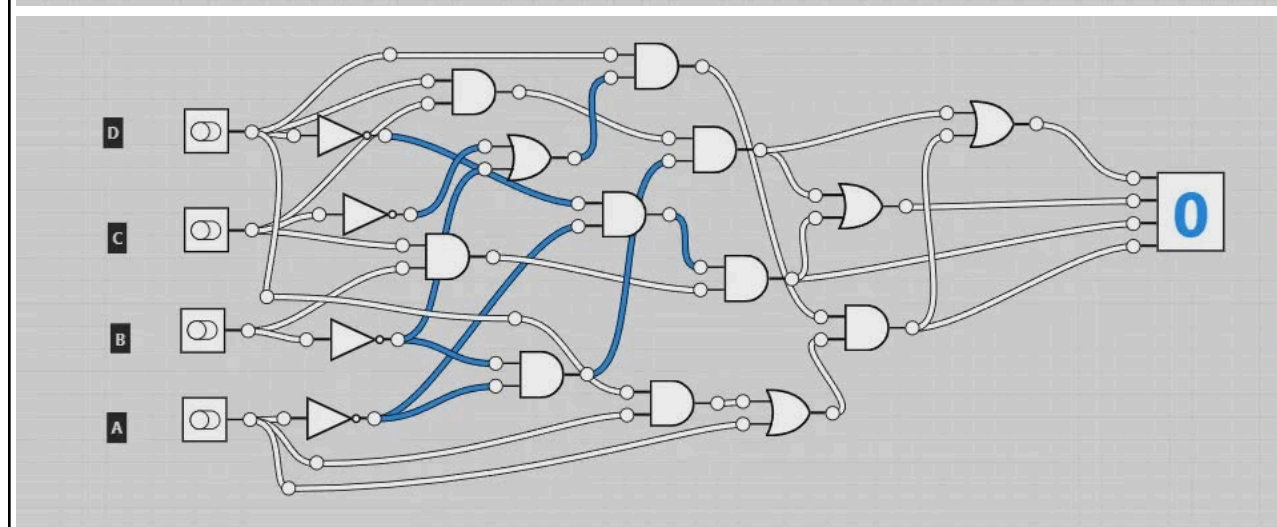
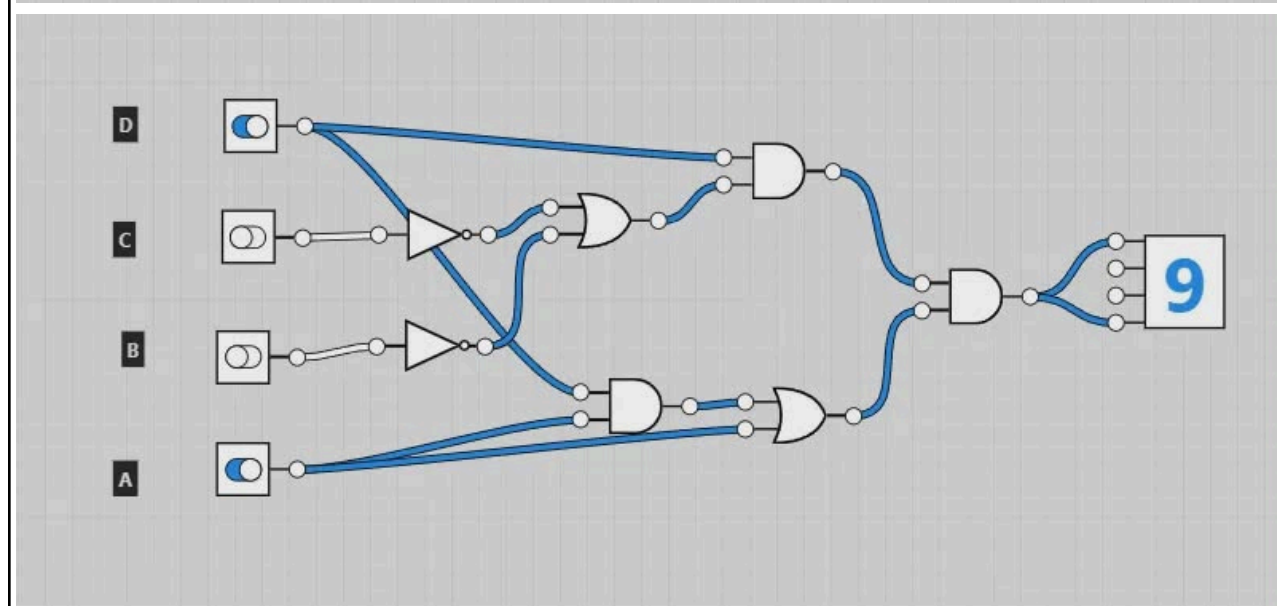
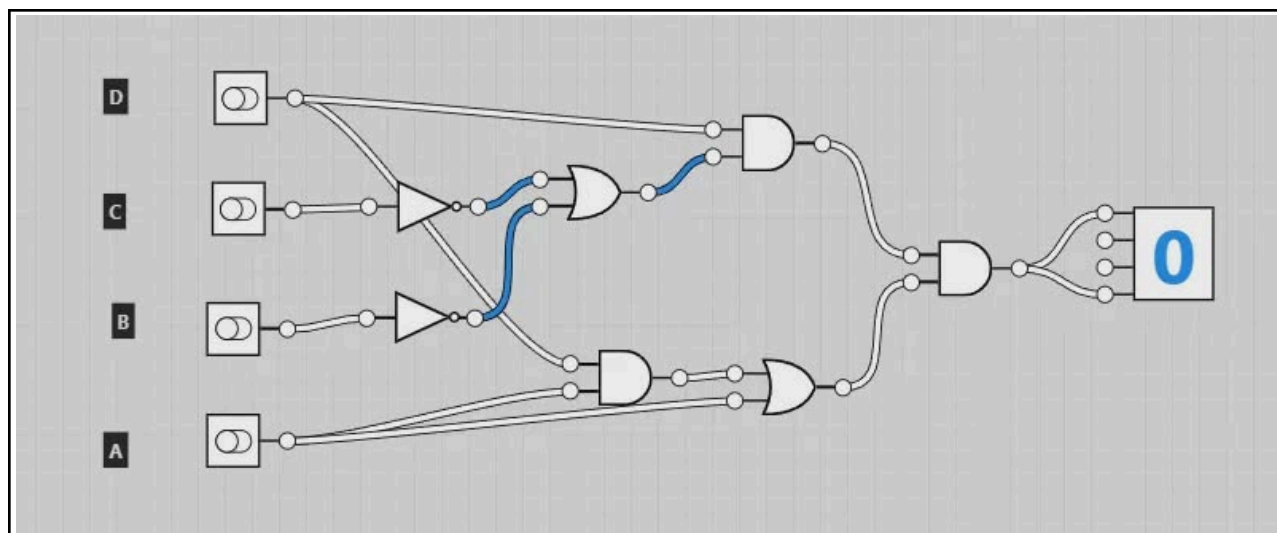
1. What have you tried to consider when you design your circuit for Problem 1? Explain.

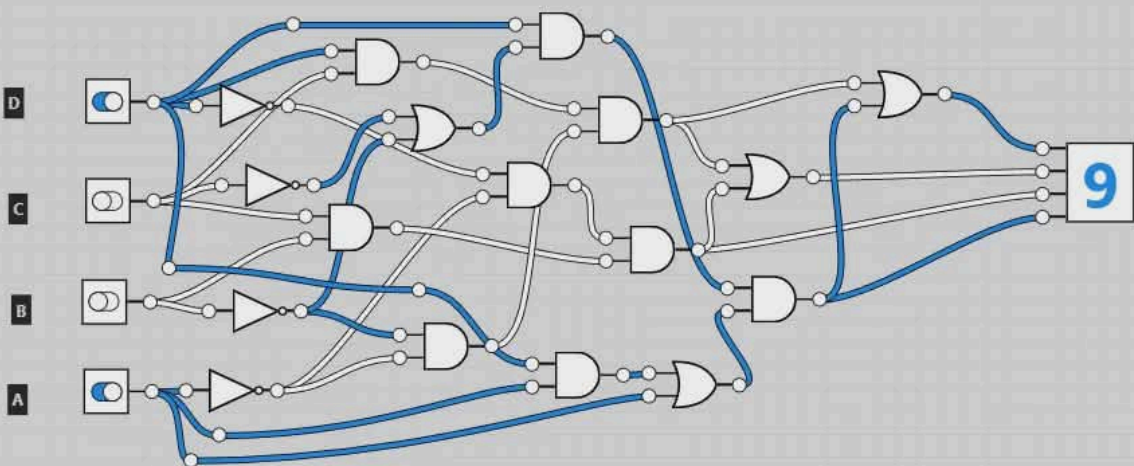
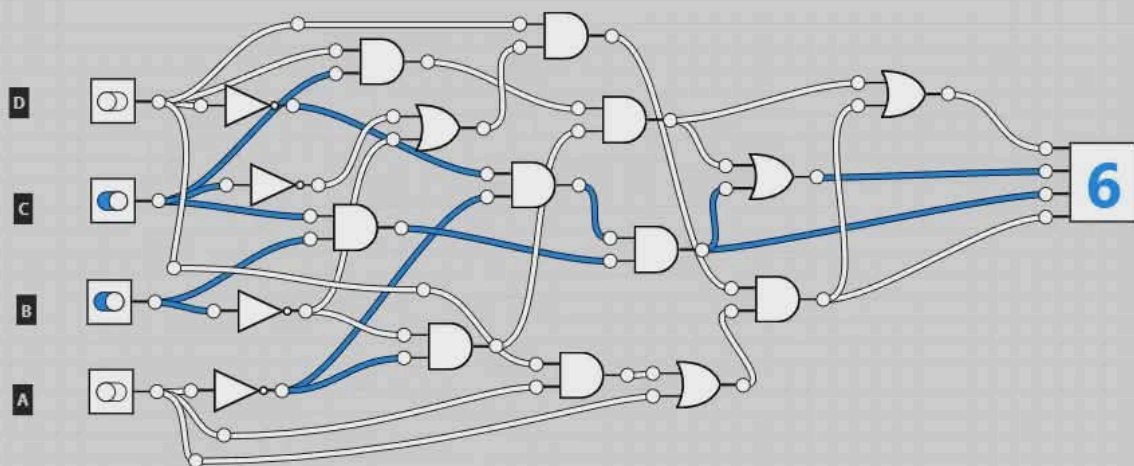
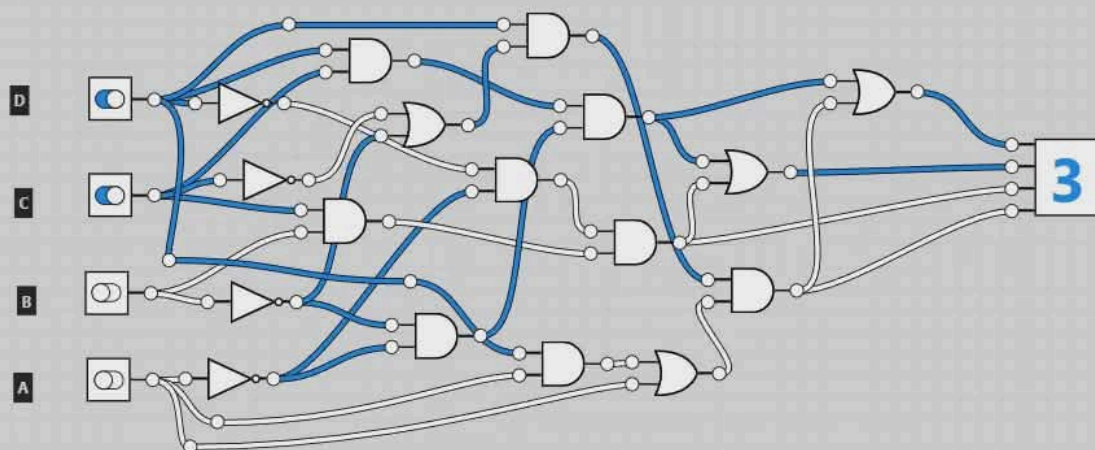
- We started by determining the necessary inputs and the outputs in order to take the problem requirements into account when creating the combinational logic circuit for the problem 1. The seven segment outputs the a-g are used to control the seven segment display, while the one control the input of y and z are utilized to represent the numbers. Then we made a truth table that displayed every possible combination of inputs and the the related outputs for the every segment. Then we simplified the boolean formulas for the each output using the K-map. This steps is important because its lower the number of the logic gates required the simplifying, improving and lowering the cost of the circuit.

2. Draw or draft a circuit diagram that will display only +3 value in 7-segment (0-3-6-9).









9. Assessment (Rubric for Laboratory Performance):

Laboratory Performance Rubric (2)   				
Criteria	Ratings			Pts
Manipulative Skills	3 pts Proficient Student always demonstrate needed skills.	2 pts Acceptable Student occasionally demonstrate needed skills.	1 pts Beginner Student do not demonstrate needed skills.	3 pts
Experimental Set - up	3 pts Proficient Student are able to set-up the material with minimum supervision.	2 pts Acceptable Student are able to set-up the materials with supervision.	1 pts Beginner Student are unable to set-up the materials.	3 pts
Process Skills	3 pts Proficient Student always demonstrate targeted process skills.	2 pts Acceptable Student occasionally demonstrate targeted process skills.	1 pts Beginner Student do not demonstrate targeted process skills.	3 pts
Application of Technical Writing Skills	3 pts Proficient Correct spelling, grammar, and sentence construction when writing the Discussion, Observation, and Conclusion are applied in the laboratory report.	2 pts Acceptable Some minor incorrect spelling, wrong grammar, and wrong sentence construction when writing the Discussion, Observation, and Conclusion are applied in the laboratory report.	1 pts Beginner Many incorrect spelling, wrong grammar, and wrong sentence construction when writing the Discussion, Observation, and Conclusion are applied in the laboratory report.	3 pts
Time Management / Conduct of Experiment	3 pts Proficient Student finish ahead of time with complete data and time to revise data.	2 pts Acceptable Student finish on time with incomplete data.	1 pts Beginner Student do not finish on time with incomplete data.	3 pts
Complete Requirements of Laboratory Report	3 pts Proficient Student submitted a complete requirements set by the teacher.	2 pts Acceptable Student submitted an almost complete Laboratory Report according to requirements set by the teacher.	1 pts Beginner Student submitted a lot of missing requirements in the Laboratory Report according to requirements set by the teacher.	3 pts
Follow the Correct Format of Laboratory Report	3 pts Proficient Student submitted a correct format of Laboratory Report according to requirements set by the teacher.	2 pts Acceptable Student submitted an almost correct format of Laboratory Report according to requirements set by the teacher.	1 pts Beginner Student did not follow at all the format of Laboratory Report according to requirements set by the teacher.	3 pts
Ability to do independent work	3 pts Proficient Student do not need to be supervised by the teacher.	2 pts Acceptable Student require occasional supervision by the teacher.	1 pts Beginner Student require supervision by the teacher.	3 pts
Total Points: 24				

